

Bayesian Structural Inference for Dynamic Crowdsourcing Contests

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Online crowdsourcing contests generate rich dynamic behavior, but their core strategic state is latent: contestants react to noisy public leaderboard feedback rather than true relative performance. We develop a tractable dynamic game with Bayesian filtering and a corresponding structural estimation framework to recover the latent competition state, effort dynamics, participant effort-cost heterogeneity, innovation volatility, and signal precision. [The model admits a closed-form leading-order Markov-perfect equilibrium in which signal precision determines how widely prize incentives extend across public performance gaps, enabling Bayesian estimation in a high-frequency dynamic setting.](#) For validation, we use a path-wise holdout design: private-leaderboard path values are excluded from the estimation likelihood and subsequently used for diagnostic tests, although final private ranks are used to define the focal rivalry. Contestants do not observe these private-leaderboard path values during the contest, but they are generated from the same submissions and scoring rule as the public leaderboard. Using data from 73 Kaggle contests, we find that contests with higher estimated innovation risk have more volatile private-leaderboard paths. For signal precision, the evidence is weaker but directionally consistent: higher estimated precision is associated with smaller public-private leaderboard disagreement. [These results provide diagnostic evidence on the estimated primitives and illustrate how the estimated model can be used to study prize levels, contest duration, and public-feedback precision. Under a calibrated score-quality mapping that holds the initial true-output environment fixed, the leading-order design analysis shows that perfect precision is optimal in a close contest, whereas finite precision can sustain incentives when the initial public gap is sufficiently large.](#)

Key words: Bayesian Statistics, Structural Estimation, Crowdsourcing Contest, Dynamic Contest

1. Introduction

Crowdsourcing contests are widely used by firms to source solutions to complex problems from external contributors. Platforms such as Kaggle¹, Topcoder², DrivenData³, and AICrowd⁴ host thousands of these contests, attracting data scientists, engineers, and developers to compete by submitting solutions to real-world problems. On Kaggle, for instance, a typical contest spans several weeks, during which participants compete for monetary prizes by repeatedly submitting solutions, observing their relative ranking on a public leaderboard, and adjusting their strategies in response to evolving feedback and the observed actions of competitors.

The central empirical challenge in these online crowdsourcing contests is that the true competitive state is unobserved during the contest. Participants observe noisy public leaderboard updates rather than true relative performance, and they choose effort based on filtered beliefs rather than on the latent state itself. The econometrician faces an analogous partial-observability problem during estimation, making it difficult to recover strategic behavior and incentives from contest-phase data alone.

Motivated by this identification problem, we develop a dynamic contest model and structural Bayesian estimation framework tailored to settings such as Kaggle, where contestants choose continuous effort while learning from noisy public leaderboard signals. Applying the framework to the Meta-Kaggle dataset (Risdal and Bozsolik 2022), we recover latent competition states and structural primitives from contest-phase public information and evaluate them using post-contest private-leaderboard paths excluded from the likelihood.

To illustrate the framework in greater detail, Figure 1 depicts the key components and their interactions in a typical Kaggle contest. One notable feature is the existence of two leaderboards: a *private* leaderboard, which is hidden until the contest ends and ultimately determines the final ranking and prize allocation, and a *public* leaderboard, which is visible to all participants during the contest.⁵ Leaderboard displays are contestant-specific and update asynchronously. After observing the public leaderboard, contestants update their beliefs and form a perceived state of the competition. Based on these beliefs, contestants choose effort according to the model’s Markov perfect equilibrium (MPE) strategies. These effort levels shape submission intensity. Each submission generates a new score for the submitting contestant, while the displayed score of the opponent remains

¹ Kaggle: <https://www.kaggle.com>

² Topcoder: <https://www.topcoder.com>

³ DrivenData: <https://www.drivendata.org>

⁴ AICrowd: <https://www.aicrowd.com>

⁵ This design serves two purposes. First, the public leaderboard provides participants with real-time feedback. Second, by keeping the private leaderboard concealed, the platform discourages overfitting to the public metric and preserves uncertainty, thereby sustaining competitive tension throughout the contest.

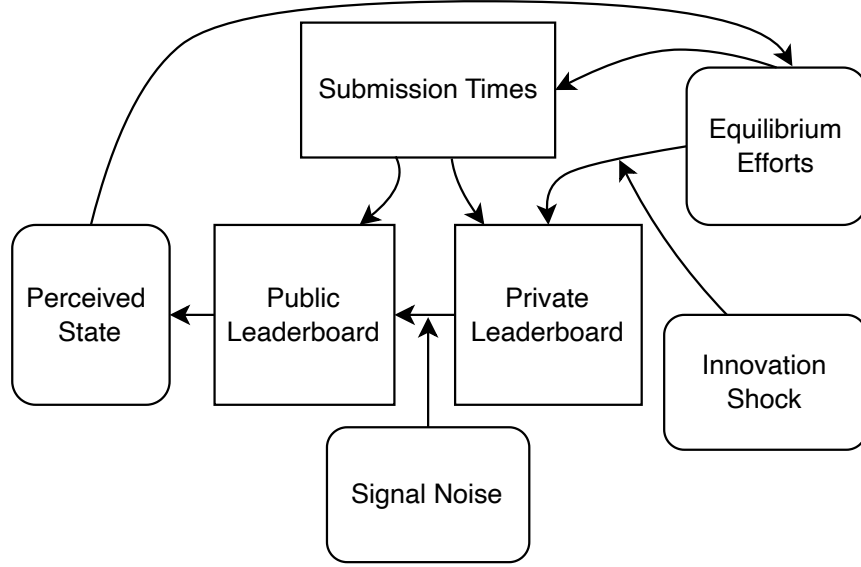


Figure 1 Structure of a Typical Kaggle Contest

Note: This figure illustrates the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles and rounded rectangles represent observed data and latent states, respectively. The private leaderboard is visible only to the contest organizer during the contest and is revealed to all participants only after it ends. During the contest period, contestants can observe only the submission times of each player and the public leaderboard.

at its most recent submitted value. The private leaderboard, in particular, evolves according to the real-time effort level and an innovation shock.

We estimate the structural parameters of the model using a Bayesian approach, based on contestants' submission times and the observed *public* leaderboard trajectories from the Meta-Kaggle database. The estimation relies solely on information available to participants during the contest and leverages the equilibrium structure implied by the dynamic model. [This procedure recovers latent competition states, effort dynamics, participant cost parameters, innovation uncertainty, and the precision of the public measurement process.](#)

To assess whether the estimated model captures meaningful contest primitives, we use private-leaderboard paths to evaluate two predictions illustrated in Figure 1: higher estimated innovation risk should be associated with greater private-path volatility, whereas higher estimated signal precision should be associated with smaller public-private leaderboard gaps. Although private-leaderboard path values are excluded from estimation, final private ranks are used to select the focal pair. We therefore interpret the validation exercises as path-wise holdout diagnostics conditional on ex post pair selection. To strengthen the credibility of these diagnostics, we also conduct time-split exercises that relate primitives estimated from early public contest data to private-path outcomes

in the later contest period, thereby reducing the concern that the full-path relationships merely reflect common path-level variation.

Empirically, we carefully select 73 high-quality contests from the Meta-Kaggle database and preprocess the data to ensure consistency across estimation methods. Conditional on the focal-pair selection rule, we estimate contest-specific innovation risk and signal precision using only public contest-phase information. The validation results show that structurally estimated innovation risk is positively associated with private-leaderboard volatility, including in a time-split diagnostic. For signal precision, the evidence is weaker but directionally consistent: higher estimated signal precision is associated with smaller public-private leaderboard disagreement, and the positive relationship between structural and benchmark precision estimates also appears under an alternative focal-pair specification.

This paper makes two main contributions. First, we develop a dynamic game-theoretic model and a Bayesian structural framework for contests with noisy public feedback and latent performance dynamics. Relative to contest models centered on discrete submit-or-stop decisions, our approach models continuous effort accumulation and makes belief updating from noisy signals structurally central. Building on the equilibrium structure of Ryvkin (2022), we model the winner as determined by the private leaderboard, hidden until the contest ends, so that the informativeness of public feedback shapes equilibrium effort incentives, while retaining a closed-form equilibrium for structural estimation. Second, we use private-leaderboard path values, which are excluded from the likelihood, to construct post-estimation validation targets. Because final private ranks enter focal-pair selection, we interpret these exercises as path-wise holdout diagnostics conditional on ex post pair selection. This approach allows us to assess whether the estimated structure captures innovation uncertainty and signal precision rather than merely fitting observed contest-phase behavior. The results provide post-estimation diagnostic evidence on the recovered contest primitives and highlight the value of the model for mechanism evaluation and platform design. Taken together, these contributions advance our understanding of online competitive dynamics and provide a foundation for future empirical work on digital contest design.

1.1. Related Literature

This study is most closely related to the following strands of literature.

Contest Design. A rich literature on *static* contest design (Ales et al. 2017) investigates how to structure competitive environments to elicit performance, typically through a contest success function mapping effort to winning probabilities (Skaperdas 1996).⁶ Because our setting is dynamic with repeated feedback, this static branch serves mainly as background for the mechanism-design

⁶ There are exceptions, such as beauty or barrier contests.

questions we build on, including prize allocation, participation rules, and information disclosure (Moldovanu and Sela 2001, Ales et al. 2021, Bergemann et al. 2022).

While these studies offer valuable insights into incentive design, they largely overlook how participants dynamically adjust their efforts in response to evolving competition. In contrast, this paper focuses on *dynamic* contests, where strategic decisions unfold over time. Two main modeling approaches have been used to capture such dynamic strategic interactions:

One strand of the literature models dynamic contests as multistage processes, where participants exert effort over time while facing uncertainty about intermediate progress. This structure facilitates the analysis of the mid-contest policy instruments available to the contest designer, such as feedback disclosure and screening mechanisms. Specifically, in a typical two-stage setting (Aoyagi 2010, Mihm and Schlapp 2019), the initial stage captures either feasibility uncertainty or foundational experimentation, with progress revealed through milestone feedback provided by the contest organizer. Participants exert effort in both stages, and the final output reflects the cumulative impact of early performance and subsequent improvement. Similarly, Bimpikis et al. (2019) study a two-stage model in which agents exert continuous effort to complete Stage A (potentially infeasible) and Stage B (feasible), with breakthroughs governed by Poisson processes. In all these models, the contest designer may strategically disclose interim feedback. Another example is Khorasani et al. (2023), who analyze a two-stage contest in which solvers invest exploratory effort to generate viable ideas, with only those passing an imperfect screening mechanism proceeding to execution.

The second approach, more closely related to our work, treats output as a stochastic process whose drift depends on the participant’s effort. This line of research originates from the tug-of-war framework, first formalized by Harris and Vickers (1987) as a one-dimensional simplification of a multi-stage R&D race. Subsequent studies build on this foundation by modeling each player’s progress as a Brownian motion. Budd et al. (1993), for example, analyze a duopoly innovation race where the state evolves as a Brownian motion driven by the difference in effort, and characterize an approximate equilibrium. Moscarini and Smith (2011) advance this approach by modeling the game state as the output gap between players and derive closed-form equilibrium strategies under pure-strategy play. Ryvkin (2022) adapt their framework to a dynamic contest with a fixed deadline, providing a closed-form solution for the Markov perfect equilibrium.

The precision of public feedback is itself a design instrument in data-science contests. Huang et al. (2024) study the training–test split, showing that more informative interim feedback makes the ranking statistically more reliable but also gives contestants the strongest incentive to submit high-variance, suboptimal models; evidence from recent Kaggle contests is consistent with this incentive effect. In contrast, our model features a different channel: more precise signals improve belief filtering and make the public ranking a more reliable predictor of the private winner.

This paper studies online crowdsourcing contests, which typically feature fixed deadlines, repeated submissions, and public real-time feedback, all governed by platform-specific rules and evaluation metrics. Hence, our analysis builds on the model by Ryvkin (2022), whose simplifications yield a closed-form equilibrium that makes structural estimation in a dynamic setting analytically and computationally tractable. We depart from their full-information setting by introducing a noisy public signal and Bayesian filtering, thereby bringing partial observability into a tractable dynamic contest model.

Structural Estimation of Contests. Beyond theoretical modeling, a growing body of research employs structural estimation techniques to recover key parameters of contest environments using observed data.

In the structural estimation of static contests, when a contest success function (CSF) is explicitly specified, the primary goal is to estimate its functional form or key parameters (Hwang 2012, Kang 2016, Huang and He 2021). In contrast, for contests without a well-defined or smooth CSF, such as beauty contests, researchers typically rely on equilibrium conditions, most notably the first-order condition. For example, Yoganarasimhan (2016) proposes a two-step semiparametric estimator, resembling the EM algorithm, to recover both unobserved auction heterogeneity and bidder cost distributions. Similarly, in auction-like models (Guerre et al. 2000, Shakhgildyan 2022), nonparametric estimation of the underlying valuation structure often serves as a first step, providing the foundation for subsequent structural estimation.

Most dynamic structural estimation exercises rely on the Markov Perfect Equilibrium (MPE) of a dynamic game. However, when the model is calibrated to closely fit real-world data, the MPE typically lacks a closed-form solution. For example, Jiang et al. (2022) develop a finite-horizon dynamic game with discrete time, state, and action spaces to model creators’ strategic behavior in response to feedback during online logo design contests. Using data from 810 contests, they implement a two-step maximum likelihood estimation (MLE) procedure under the assumption of ex-ante homogeneous contests and players: the first step estimates action-specific state transition probabilities, while the second recovers utility parameters.

Specific to online crowdsourcing environments, Zhang et al. (2019) build a dynamic structural model to study how participants in crowdsourcing contests learn and choose contests over time. Their model shows that competing against superstars accelerates learning by providing more informative feedback. Using Bayesian updating and a hierarchical model, they estimate how individuals trade off monetary rewards, learning opportunities, and participation costs, based on data from Topcoder.

Most closely related to our work, Lemus and Marshall (2021) structurally estimate contest behavior using Kaggle data. They propose a finite-horizon dynamic game in which, after each

submission, players decide whether to continue or exit, essentially solving an optimal stopping problem. Players can only make this decision immediately after completing a submission, which arrives at random intervals and incurs a privately drawn cost. Their estimation proceeds in two stages: the first uses maximum likelihood and the EM algorithm to estimate primitive parameters such as the distribution of player types and scores; the second applies a conditional choice probability (CCP)-based estimator to identify the cost distribution. Our model differs in two key respects: (1) we treat submissions as the outcome of accumulated effort rather than as discrete costly actions; and (2) we explicitly model belief updating from noisy signals as a structural channel that shapes both behavior and identification, whereas their CCP-based approach relies on a coarser behavioral approximation.

The structure of the paper is as follows: Section 2 introduces our theoretical model of online crowdsourcing contests. Section 3 presents the Bayesian estimation framework. In Section 4, we describe the data-generating process and apply the Bayesian framework to synthetic data. Section 5 applies the framework to real-world Kaggle data and evaluates the estimated primitives using private-leaderboard path values excluded from the likelihood. A replication package (code and implementation details) will be provided with the paper.

2. The Model

Two players, i and j , compete for a prize θ in a contest. The winner receives the prize and the loser receives zero. The contest starts at time zero. At each time $t \geq 0$, player i chooses an effort level $q_{i,t}$ and incurs quadratic cost $C_i(q_{i,t}) = c_i q_{i,t}^2 / 2$, where lower c_i corresponds to higher ability. Let y_t denote the *latent private-score gap* between players i and j at time t , evolving as

$$dy_t = (q_{i,t} - q_{j,t})dt + \sigma dW_t \quad (1)$$

where W_t is a Brownian motion and $\sigma > 0$ measures the innovation risk. When σ is large and the relationship between effort and output is highly uncertain, luck becomes the dominant factor. Conversely, when σ is small and output closely reflects effort, the importance of effort is amplified.

The actual contest is equipped with a submission system in which contestants submit solutions asynchronously over time. Each submission generates new public and private scores for the submitting contestant, while the opponent's most recently submitted scores are carried forward when constructing the corresponding leaderboard gaps. For tractability, we approximate this irregular sequence of submission-triggered updates by a continuous-time public signal of the latent private-score gap y_t . The signal evolves according to

$$dZ_t = y_t dt + \frac{dB_t}{\sqrt{\lambda}} \quad (2)$$

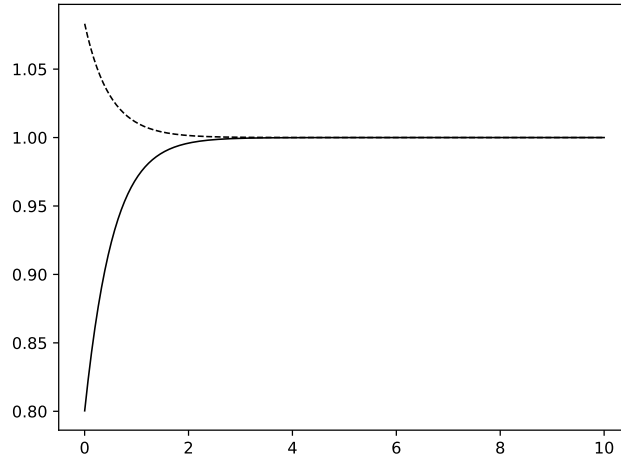


Figure 2 The evolution of S_t in time t , given that $\lambda = 1$ and $\sigma = 1$.

Note: The plot shows the conditional variance S_t of the filtering error (i.e., the uncertainty about the private-score gap) for $\lambda = \sigma = 1$.

where B_t is a standard Brownian motion independent of W_t , and λ controls signal precision. A larger λ implies less noise and more informative feedback; in the empirical mapping, this corresponds to a more informative public leaderboard signal about latent relative performance. This continuous-time signal is an idealized analogue of the stream of public leaderboard updates observed after submissions.

We study public Markov equilibria under complete information. Let I_t denote the common history of public signals and past efforts, and define the posterior mean and variance by $\tilde{y}_t := E(y_t | I_t)$ and $S_t := E[(y_t - \tilde{y}_t)^2 | I_t]$. Initially, $y_0 | I_0 \sim \mathcal{N}(\tilde{y}_0, S_0)$, where $\tilde{y}_0$ is publicly known and $S_0 > 0$ is fixed independently of λ . Under standard linear-Gaussian assumptions, the Kalman-Bucy filter (Chapter 1.2 of Bensoussan 1992; see also Frogerais et al. 2012, Barrau and Bonnabel 2017) implies

$$d\tilde{y}_t = (q_{i,t} - q_{j,t})dt + \lambda S_t(dZ_t - \tilde{y}_t dt) \quad (3)$$

$$\frac{dS_t}{dt} = \sigma^2 - \lambda S_t^2 \quad (4)$$

Hence, the conditional distribution satisfies $y_t | I_t \sim \mathcal{N}(\tilde{y}_t, S_t)$, so the posterior state is fully characterized by $(\tilde{y}_t, S_t)$. The steady-state value associated with equation (4) is $\bar{S} \equiv \sigma/\sqrt{\lambda}$. Given S_0 , equation (4) determines a deterministic path S_t , so $(\tilde{y}_t, t)$ remains a sufficient strategic state. Appendix B.1 reports the solution.

Following Ryvkin (2022), we consider a dynamic contest with a fixed deadline. Suppose the contest is terminated at time $T > 0$. Let $\tilde{y}_T$ denote player i 's perceived terminal performance advantage over player j . The sign of y_T determines the winner. Since $y_T | I_T \sim \mathcal{N}(\tilde{y}_T, S_T)$, the

expected terminal prizes are $G_i(\tilde{y}_T) = \theta\Phi(\tilde{y}_T/\sqrt{S_T})$ and $G_j(\tilde{y}_T) = \theta\Phi(-\tilde{y}_T/\sqrt{S_T})$. Taking player j 's effort path as given, player i 's value at $(\tilde{y}, t)$ is

$$V^i(\tilde{y}, t) = \sup_{\{q_{i,\tau} \geq 0\}_{\tau \in [t, T)}} \mathbb{E} \left[G_i(\tilde{y}_T) - \int_t^T C_i(q_{i,\tau}) d\tau \mid \tilde{y}_t = \tilde{y} \right],$$

The optimization is subject to equations (3)–(4). Player j 's problem is obtained by replacing $\tilde{y}$ with $-\tilde{y}$ and exchanging the player labels and cost parameters. The corresponding Hamilton-Jacobi-Bellman (HJB) equation for player i is

$$0 = \max_{q_{i,t} \geq 0} \left[-\frac{c_i q_{i,t}^2}{2} + V_y^i \cdot (q_{i,t} - q_{j,t}) + V_t^i + \frac{V_{yy}^i}{2} \lambda S_t^2 \right].$$

Assuming an interior solution, substituting the first-order conditions $q_{i,t} = V_y^i/c_i$ and $q_{j,t} = -V_y^j/c_j$ yields the system

$$\begin{aligned} \frac{1}{2c_i} (V_y^i)^2 + \frac{1}{c_j} V_y^i V_y^j + V_t^i + V_{yy}^i \frac{\lambda S_t^2}{2} &= 0 \\ \frac{1}{2c_j} (V_y^j)^2 + \frac{1}{c_i} V_y^j V_y^i + V_t^j + V_{yy}^j \frac{\lambda S_t^2}{2} &= 0 \end{aligned}$$

subject to boundary conditions $V^i(-\infty, t) = 0$, $V^i(+\infty, t) = \theta$, $V^j(-\infty, t) = \theta$, $V^j(+\infty, t) = 0$, and terminal conditions $V^i(\tilde{y}_T, T) = G_i(\tilde{y}_T)$ and $V^j(\tilde{y}_T, T) = G_j(\tilde{y}_T)$. The Nash equilibrium is summarized in the following theorem.

THEOREM 1 (Leading-Order Equilibrium Effort). *Suppose an interior Markov-perfect equilibrium admits a regular expansion in $\bar{w}$, uniform for the HJB derivatives. Let $w_k := \theta/(\sigma^2 c_k)$ and $\bar{w} := \max\{w_i, w_j\}$. For fixed finite $\lambda > 0$ and $S_0 > 0$ independent of λ , as $\bar{w} \downarrow 0$ equilibrium effort satisfies*

$$m_k(\tilde{y}_t, t) = \frac{\sigma^2}{\sqrt{Q_t}} \left\{ \frac{w_k}{\sqrt{2\pi}} \exp \left[-\frac{\tilde{y}_t^2}{2Q_t} \right] + O(\bar{w}^2) \right\}, \quad Q_t := \sigma^2(T - t) + S_t, \quad (5)$$

for $k \in \{i, j\}$ and $t \in [0, T]$.

Integrating equation (4) yields

$$S_T + \int_t^T \lambda S_s^2 ds = S_t + \sigma^2(T - t) = Q_t,$$

which gives the closed-form kernel in equation (5).

2.1. Implications for Contest Design

Let x_t^k denote contestant k 's private performance score, normalized by

$$\frac{x_0^i + x_0^j}{2} = 0, \quad y_0 := x_0^i - x_0^j, \quad y_0 \mid I_0 \sim \mathcal{N}(\tilde{y}_0, S_0). \quad (6)$$

Individual scores evolve according to

$$dx_t^k = q_{k,t}dt + \frac{\sigma}{\sqrt{2}}dW_t^k, \quad k \in \{i, j\}, \quad (7)$$

where W^i and W^j are independent and $q_{k,t} = m_k(\tilde{y}_t, t)$ in equilibrium. With $W_t = (W_t^i - W_t^j)/\sqrt{2}$, the gap obeys equation (1). The organizer values the best terminal score,

$$\mathcal{B}(\tilde{y}_0, \theta, T; S_0) := \mathbb{E}[\max\{x_T^i, x_T^j\} \mid I_0].$$

LEMMA 1 (Total Expected Effort). *Suppose $c_i = c_j = c$, and let $M(\tilde{y}_t, t) := \mathbb{E}[\int_t^T (m_i(\tilde{y}_s, s) + m_j(\tilde{y}_s, s))ds \mid \tilde{y}_t]$. As $w = \theta/(\sigma^2 c) \downarrow 0$,*

$$M(\tilde{y}_t, t) = \frac{2\sigma^2(T-t)}{\sqrt{Q_t}} \left\{ \frac{w}{\sqrt{2\pi}} \exp\left[-\frac{\tilde{y}_t^2}{2Q_t}\right] + O(w^2) \right\}, \quad (8)$$

where $Q_t = \sigma^2(T-t) + S_t$.

LEMMA 2 (Expected Best Output). *Under equation (6) and symmetric costs, as $w = \theta/(\sigma^2 c) \downarrow 0$, expected best output satisfies $\mathcal{B} = \bar{\mathcal{B}} + O(w^2 \sigma^2 T / \sqrt{Q_0})$, where $Q_0 = S_0 + \sigma^2 T$ and*

$$\begin{aligned} \bar{\mathcal{B}}(\tilde{y}_0, \theta, T; S_0) := & \sqrt{\frac{Q_0}{2\pi}} \exp\left(-\frac{\tilde{y}_0^2}{2Q_0}\right) + |\tilde{y}_0| \left[\Phi\left(\frac{|\tilde{y}_0|}{\sqrt{Q_0}}\right) - \frac{1}{2} \right] \\ & + \frac{\theta T}{c\sqrt{2\pi}Q_0} \exp\left[-\frac{\tilde{y}_0^2}{2Q_0}\right]. \end{aligned} \quad (9)$$

The first two terms give expected best output without effort, and the final term equals one half of leading-order total effort. Since $Q_0 = S_0 + \sigma^2 T$, the leading-order objective is independent of λ ; comparing precision regimes therefore requires higher-order terms or an additional benefit or cost of precision.

3. Estimation Framework

This section presents the data-generating process and the Bayesian estimation procedure. We first connect the theoretical model to the observable contest data and then describe the likelihood construction and prior specification.

3.1. Data-Generating Process

For each Kaggle contest, the observable data from Meta-Kaggle can be grouped into three components.

The first component contains contest-level institutional features: contest duration, prize structure, and information disclosure rules. To match the two-player winner-take-all structure of the model, we restrict the empirical application to contests where a dominant top rivalry is plausible

and monetary prize incentives can be mapped to an effective top-two winner-gain structure. The effective prize θ is observed rather than estimated. We express monetary payoffs in units of one hundred thousand U.S. dollars and use this unit for every contest.

The second component records submission events of player i , denoted by $\{\hat{t}_k^i\}_{k=1}^{N_i}$, where N_i is the total number of submissions. Conditional on latent effort paths, we model the submissions of players i and j as two inhomogeneous Poisson processes with intensities $\tau_i(t)$ and $\tau_j(t)$. These processes are conditionally independent given the latent effort paths. For any interval $\mathcal{S} \subset \mathcal{T}$, the expected number of submissions by player i is $\int_{\mathcal{S}} \tau_i(s) ds$. To link observed submission timing to latent equilibrium effort, we assume for tractability that

$$\tau_i(t) = r \cdot m_i(\tilde{y}_t, t) \quad (10)$$

where $r > 0$ converts model effort into submission intensity; the same relationship applies to player j . This proportional relationship is an empirical measurement assumption rather than a result derived from the theoretical game. In estimation, r is treated as an auxiliary scale parameter rather than a contest primitive.

The third component contains *public* and *private* leaderboard series, denoted by $\hat{x}_t^{i(j)}$ and $x_t^{i(j)}$. In Kaggle contests, public scores are computed from a released subset, while private scores are computed from withheld data and revealed only after the contest ends. We define the latent contemporaneous gap as $y_t := x_t^i - x_t^j$ and the displayed public gap as $\hat{y}_t := \hat{x}_t^i - \hat{x}_t^j$, where each displayed score equals the contestant's most recently submitted public score. Because submissions are asynchronous, $\hat{y}_t$ need not compare contemporaneous submissions by the two contestants. The public gap is the empirical counterpart of the noisy signal in (2), observed through discrete leaderboard updates. We write this empirical readout as

$$dZ_t = \hat{y}_t dt, \quad (11)$$

which should be interpreted as an observation convention linking the public leaderboard gap to the signal path, not as a second structural definition of Z_t . Combining (2) and (11) gives the formal white-noise representation

$$\hat{y}_t = y_t + \frac{1}{\sqrt{\lambda}} \xi_t, \quad \xi_t \equiv \frac{dB_t}{dt}, \quad (12)$$

where ξ_t denotes Gaussian white noise in the generalized-process sense. Because white noise is not pointwise observed, equation (12) is not a pointwise sampling equation. In the empirical data, the public leaderboard is sampled only at submission-triggered update times. At an empirical update time t_k , averaging the white-noise representation over the observed readout interval $h_k = t_k - t_{k-1}$

yields a finite-variance noise term $\varepsilon_k \sim \mathcal{N}(0, 1/(\lambda h_k))$. Approximating the corresponding readout of the latent gap by its endpoint value and substituting the latent state dynamics in (1) gives

$$\hat{y}_{t_k} \approx y_0 + \int_0^{t_k} [m_i(\tilde{y}_s, s) - m_j(\tilde{y}_s, s)] ds + \sigma W_{t_k} + \varepsilon_k, \quad (13)$$

for update times t_k . Here, σW_{t_k} captures accumulated innovation noise, while ε_k captures the public-signal noise associated with the leaderboard update. Thus, larger λ implies more precise public feedback. The realized initial gap is not observed during the contest and does not enter the likelihood as observed data. Consistent with the theoretical steady-state specification, we instead treat it as a latent draw from

$$y_0 \sim \mathcal{N}(\mu_0, \bar{S}(\lambda)), \quad \bar{S}(\lambda) = \frac{\sigma}{\sqrt{\lambda}}, \quad (14)$$

and integrate over y_0 in the likelihood. The parameter μ_0 represents the initially perceived gap, whereas y_0 is its unobserved realization.

The theoretical model uses a continuous public signal and therefore admits the continuous-time filtering equation in (3). The empirical leaderboard, however, is observed only at asynchronous submission times. An exact empirical implementation would therefore require a hybrid filter with continuous prediction between submissions and discrete measurement updates.

To preserve the tractable one-dimensional state representation used in the equilibrium analysis, we instead employ a continuous-update approximation. Specifically, we carry forward the most recently observed public gap and evolve the perceived gap according to

$$d\tilde{y}_t = [m_i(\tilde{y}_t, t) - m_j(\tilde{y}_t, t)] dt + \sqrt{\lambda} \sigma (\hat{y}_t - \tilde{y}_t) dt \quad (15)$$

with initial condition $\tilde{y}_0 = \mu_0$. This approximation may overstate learning during long intervals without submissions. The resulting estimates should therefore be interpreted as conditional on the continuous-update approximation rather than as estimates from an exact discrete-observation filter.

We denote the set of estimated parameters by $\Theta := \{c_i, c_j, \sigma, \lambda, \mu_0, r\}$. The scale parameter r is estimated as an auxiliary measurement parameter that maps latent effort into submission intensity. The precision parameter λ enters both the observation system and the equilibrium effort rule. Once Θ is specified, equations (1), (13), and (15) define the working empirical system, while realized trajectories remain stochastic.

3.2. Modeling Considerations

This subsection discusses three practical modeling concerns that are important for interpretation: strategic submission timing, the two-player approximation, and the normality assumption. The goal is to clarify the scope of the baseline framework and connect each concern to the robustness analyses that follow.

Strategic submission timing. Beyond the uncertainty caused by partial data disclosure, an additional source of signal noise stems from the timing of leaderboard updates: rankings are only refreshed upon new submissions, rendering the leaderboard uninformative during periods of inactivity. This lag can bias inference if participants deliberately time or withhold submissions.

Our baseline observation equation assumes submissions are driven by latent effort and does not explicitly model strategic withholding. Strategic submission timing is an important concern in the literature. For example, Lemus and Marshall (2021) adopt a similar simplification by excluding such behavior from their model. Our estimates should therefore be interpreted for contests where active updating behavior dominates withholding behavior. As a partial mitigation, we require a minimum number of submissions when selecting representative contestants (Section 5.1), which reduces cases where sparse strategic timing dominates observed dynamics.

Selection of Two Players. Another concern is the two-player approximation. Real contests involve many participants, so reducing the environment to two players changes strategic externalities and information structure. Our interpretation is that the model isolates the dominant top rivalry in contests where competition is concentrated near the frontier. In Section 6.1, we conduct a series of robustness checks. By varying the focal-pair definition, we assess whether the validation results depend on a particular choice of representative rivalry.

Gaussian approximation. The model relies on Gaussian signal and innovation shocks. Mapping bounded leaderboard scores into the model’s real-valued state space therefore requires both a transformation and a maintained distributional approximation. The empirical transformation is defined in Section 5.1. Rescaling the transformed scores changes their units but does not determine whether their distribution is Gaussian. We therefore treat Gaussianity as a modeling assumption rather than an empirically established property of every contest.

3.3. Bayesian Estimation Framework

Our objective is to develop a Bayesian framework for inferring the unknown parameter set $\Theta := \{c_i, c_j, \sigma, \lambda, \mu_0, r\}$ using the data introduced above.

Conditional on the selected focal pair, the estimation likelihood uses only contest-phase submission timing and public leaderboard updates.

For numerical integration of the effort and filtering equations, we use a uniform computational grid of length Δ_{num} . This numerical step size does not determine the observation-noise variance. By

definition, the likelihood function corresponding to the submission times of contestant i , $\{t_k^i\}_{k=1}^{N_i}$, is given as follows (a symmetric formulation applies to contestant j):

$$p\left(\{t_k^i\}_{k=1}^{N_i} \mid \tau_i, \Theta\right) = \exp\left\{-\int_{s \in \mathcal{T}} \tau_i(s) ds\right\} \prod_{k=1}^{N_i} \tau_i(t_k^i) \quad (16)$$

where τ_i is defined in (10). The integral term is evaluated on the computational grid.

Next, the public leaderboard gap $\hat{y}_t$ enters the likelihood only when either contestant submits and receives an update. We denote these update times by the ordered sequence $(t_1, t_2, \dots, t_N)$, where $N = N_i + N_j$. Each t_k is associated with an observed public gap $\hat{y}_{t_k}$. Let $h_k = t_k - t_{k-1}$, with $t_0 = 0$, denote the elapsed calendar time associated with the k th empirical readout. The quantities h_k are determined by observed submission timing and are distinct from the numerical integration step Δ_{num} .

To make the conditioning structure explicit, we evaluate the leaderboard likelihood sequentially. Let $\mathcal{F}_{k-1}$ denote the submission and public-leaderboard history strictly before update time t_k . Conditional on $\mathcal{F}_{k-1}$ and Θ , the empirical filtering recursion generates the one-step-ahead predictive distribution

$$\hat{y}_{t_k} \mid \mathcal{F}_{k-1}, \Theta \sim \mathcal{N}(\mu_{k|k-1}, V_{k|k-1}),$$

Here, $\mu_{k|k-1}$ and $V_{k|k-1}$ denote the one-step-ahead predictive mean and variance implied by the empirical state and observation equations over $(t_{k-1}, t_k]$, conditional only on $\mathcal{F}_{k-1}$. These moments are computed before incorporating the leaderboard observation at t_k . The leaderboard likelihood is therefore

$$p\left(\{\hat{y}_{t_k}\}_{k=1}^N \mid \Theta\right) = \prod_{k=1}^N \phi(\hat{y}_{t_k} \mid \mu_{k|k-1}, V_{k|k-1}). \quad (17)$$

The observation at t_k is incorporated into the perceived-state recursion only after its conditional density has been evaluated, so it affects subsequent predicted states and effort paths but not its own one-step-ahead predictive mean. Each predictive density integrates over the initial distribution in equation (14). The parameter μ_0 directly determines the initial predictive mean and also affects subsequent perceived states and effort paths through the filtering recursion in equation (15). Accordingly, the likelihood uses the public leaderboard values associated with submission updates and preserves the temporal ordering of information.

It is worth noting that, although the private leaderboard trajectory y_t is observable ex post in Meta-Kaggle, we deliberately exclude it from the estimation likelihood. In our framework, agents form beliefs about y_t based solely on $\hat{y}_t$, and their actions are driven by these beliefs. Conditioning on the realized values of y_t in estimation would effectively bypass the agent's informational constraints and undermine the role of λ in shaping belief formation. More importantly, it would destroy the

path-wise holdout design based on private-leaderboard values. Treating y_t as latent preserves the model’s behavioral interpretation and enables post-estimation path validation, while final private ranks remain part of focal-pair selection.

To evaluate (16) and (17), we recursively construct $\tilde{y}_t$ and implied efforts (m_i, m_j) from (5) and (15). In the empirical model, m_k denotes the leading-order term in equation (5); the $O(\bar{w}^2)$ remainder is omitted. More precisely, conditional on a parameter vector, perceived states and effort paths are constructed by forward filtering, using each public update only for subsequent states. Because the effort rule in equation (5) is closed form, it can be evaluated directly within Hamiltonian Monte Carlo and the No-U-Turn Sampler (Hoffman and Gelman 2014).

In the empirical application, we set $c_i, c_j \sim \text{LogNormal}(\log 1, 0.75^2)$, $\sigma \sim \text{LogNormal}(\log 2, 0.5^2)$, $\lambda \sim \text{LogNormal}(\log 1, 0.75^2)$, $r \sim \text{LogNormal}(\log 10, 0.5^2)$, and $\mu_0 \sim \mathcal{N}(\hat{y}_{t_1}, 1)$. All positive parameters are constrained to be at least 0.01 and have no upper bound. Because the prior center for μ_0 uses the first observed public gap, this specification is an empirical-Bayes initialization. The posterior is therefore conditional on this initialization rule, while the unit prior variance leaves the initial perceived gap uncertain.

PROPOSITION 1 (Local Identification). *Suppose $\theta > 0$, the public leaderboard is observed at least three distinct times, and $c_i \neq c_j$. Then the interior parameter vector $\Theta = (c_i, c_j, \sigma, \lambda, \mu_0, r)$ is locally identified from equations (16) and (17).*

The asymmetry condition is essential: identification becomes weak as c_i approaches c_j and fails at equality. Appendix A.1 provides the proof.

4. Synthetic Experiments

Before applying our estimation procedure to real-world contest data, we evaluate its potential on synthetically generated data. These synthetic exercises serve three purposes: illustrating the data-generating mechanism, assessing finite-sample parameter recovery, and clarifying which parameters are easier or harder to recover under realistic data conditions. All experiments use the leading-order effort rule in equation (5) and draw the latent initial gap from the common initial belief.

A central challenge in generating synthetic data lies in dynamically constructing a point process that conforms to an inhomogeneous Poisson process. We first generate candidate submission events according to a homogeneous Poisson process over the whole contest duration, using a fixed high intensity $\tau^* \geq \sup_t \tau_{i(j)}(t)$. Then, we apply the classical thinning procedure (Lewis and Shedler 1979) to determine whether each candidate event is accepted or not. In plain terms, Algorithm 1 generates candidate submissions, simulates latent and perceived states, accepts submissions via thinning, updates public leaderboard observations, and produces the synthetic sample used in the subsequent Bayesian recovery exercise.

Algorithm 1 Synthetic Data Simulation**Input:** $\Delta, T, \theta, c_i, c_j, \sigma, \lambda, r, \tau^*, \mu_0$ Compute $\bar{S}(\lambda) = \sigma/\sqrt{\lambda}$ and draw $y_0 \sim \mathcal{N}(\mu_0, \bar{S}(\lambda))$.Sample candidate sets $\mathcal{C}_{i(j)}$ from Poisson process (τ^*) on $[0, T]$ For each $s \in \mathcal{C}_{i(j)}$, sample $u_s^{i(j)} \sim \mathcal{U}[0, 1]$ Sample Brownian increments (ΔW_t) and (ΔB_t) Initialize $\ell^{i(j)} = 1$ and $\hat{y}_0 = \tilde{y}_0 = \mu_0$.**for** $t = 0, \Delta, \dots, T - \Delta$ **do** $m_{i(j)}(\tilde{y}_t, t) \leftarrow (5), \tau_{i(j)}(t) \leftarrow (10)$ {Use $\theta, c_{i(j)}, \sigma, \lambda, r, T$ } $y_{t+\Delta} \leftarrow (1)$ {Use $m_i - m_j, \sigma, \Delta, \Delta W_t$ } $\tilde{y}_{t+\Delta} \leftarrow (15)$ {Use $m_i - m_j, \lambda, \sigma, \Delta$, and available $\hat{y}_t$ } $\hat{y}_{t+\Delta} \leftarrow \hat{y}_t$ **for** $s \in \mathcal{C}_{i(j)} \cap [t, t + \Delta)$ **do****if** $u_s^{i(j)} < \tau_{i(j)}(t)/\tau^*$ **then** $\hat{t}_\ell^{i(j)} \leftarrow s; \ell^{i(j)} \leftarrow \ell^{i(j)} + 1$ {Accept the submission event} $\hat{y}_{t+\Delta} \leftarrow$ discretized readout of (13) {Use $\sigma, \lambda, \Delta B_t$; stored for subsequent filtering}**end if****end for****end for****Output:** $(y_t), (\tilde{y}_t), (\hat{y}_t), (m_i, m_j)$, and accepted submission times $\{\hat{t}_k^{i(j)}\}$

As a baseline example, we consider a synthetic contest with a three-month horizon. The two participants differ slightly in effort costs, with $c_i = 1.2$ and $c_j = 1.5$. The remaining parameters are set to $\theta = 1$, $\sigma = 2.0$, $\lambda = 1.0$, and $r = 15$. These values imply $w_i \approx 0.208$ and $w_j \approx 0.167$, placing the baseline experiment in the small-incentive region used by the leading-order equilibrium.

Figure 3 shows one baseline realization. The differences among y_t , $\hat{y}_t$, and $\tilde{y}_t$ illustrate how submission-triggered noisy readouts are translated into contestants' perceived state. The realization begins with a latent gap of -1.445 and produces 30 submissions by player i and 16 by player j . Between submissions, filtering uncertainty would generally evolve over time; for tractability, however, the synthetic design follows the maintained steady-state specification used throughout the model.

The synthetic recovery exercises use simulation-specific priors, reported in the table notes, and estimate $(c_i, c_j, \sigma, \lambda, \mu_0)$ while fixing the auxiliary effort-to-submission scale r at its simulation value. When describing a parameter as weakly pinned down in these Bayesian recovery exercises, we use a simple posterior-concentration diagnostic: the posterior standard deviation remains close

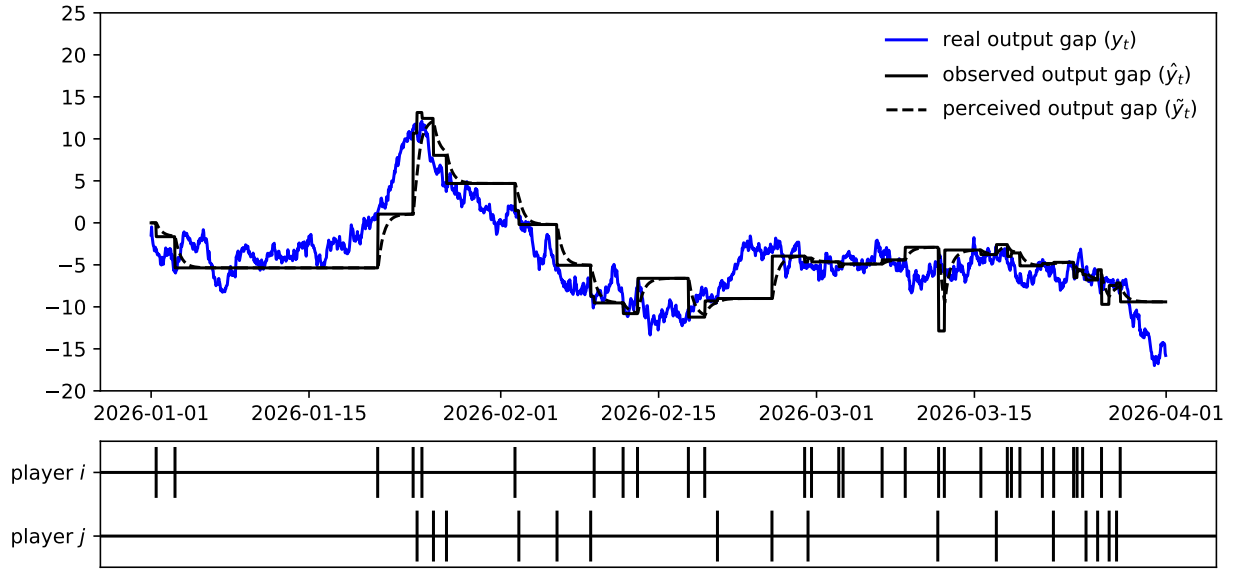


Figure 3 Synthetic Contest Data over a Three-Month Horizon

Note: The blue line represents the latent output gap y_t , the solid black line represents the public leaderboard $\hat{y}_t$ (updated only upon submissions), and the dashed black line represents the perceived output gap $\tilde{y}_t$ inferred via Bayesian filtering. The lower panel marks submission times in separate rows for players i and j . Parameters: $\theta = 1.0$, $c_i = 1.2$, $c_j = 1.5$, $\sigma = 2.0$, $\lambda = 1.0$, $r = 15$, $T = 3$ months.

to its prior standard deviation, and the credible interval shows little shrinkage relative to the prior support. For μ_0 , the prior standard deviation is approximately one, so posterior standard deviations near one indicate limited learning from the realized synthetic data.

Panel A of Table 1 presents the Bayesian inference results based on the synthetic data displayed in Figure 3. With limited data from a short contest, recovery is uneven across parameters. Recovery is most precise for c_i and σ , weaker for c_j and λ , and especially limited for μ_0 . Thus, clean computational convergence does not imply that one contest path sharply identifies every primitive. This finite-sample pattern is informative and motivates the richer-data exercises below. Panel B repeats the baseline exercise while estimating r rather than fixing it. The posterior estimates of the remaining primitives are similar to those in Panel A, so allowing r to vary does not materially change baseline recovery.

Table 1 Bayesian Estimates from Synthetic Data

Panel A. r Fixed at Its True Value						
Parameters (true val.)	c_i	c_j	σ	λ	μ_0	
	1.2	1.5	2.0	1.0	0.0	
Posterior Mean	1.162	2.077	2.263	1.363	-0.318	
Posterior SD	0.224	0.519	0.264	0.529	0.931	
Post. RMSD	0.227	0.776	0.373	0.641	0.984	
95% Interval	[0.80, 1.66]	[1.29, 3.24]	[1.79, 2.84]	[0.60, 2.67]	[-2.13, 1.54]	

Panel B. r Estimated Jointly						
Parameters (true val.)	c_i	c_j	σ	λ	μ_0	r
	1.2	1.5	2.0	1.0	0.0	15.0
Posterior Mean	1.072	1.910	2.251	1.340	-0.349	13.718
Posterior SD	0.369	0.664	0.262	0.493	0.930	4.046
Post. RMSD	0.391	0.780	0.363	0.598	0.993	4.244
95% Interval	[0.53, 1.99]	[0.90, 3.51]	[1.80, 2.82]	[0.59, 2.55]	[-2.18, 1.44]	[7.49, 22.88]

Note: This table reports posterior means, posterior standard deviations, posterior root mean squared deviations (Post. RMSD), and 95% credible intervals for the synthetic realization in Figure 3, in which players i and j submit 30 and 16 times. Panel A fixes r at its simulation value; Panel B estimates it jointly with $(c_i, c_j, \sigma, \lambda, \mu_0)$ using the prior $r \sim \text{Lognormal}(\log 15, 0.35^2)$. Both panels use $c_i, c_j \sim \text{Lognormal}(\log 1, 0.75^2)$, $\sigma \sim \text{Lognormal}(\log 2, 0.5^2)$, $\lambda \sim \text{Lognormal}(0, 0.75^2)$, and $\mu_0 \sim \mathcal{N}(0, 1)$, with supports $[0.1, 5]$, $[0.5, 10]$, $[10^{-6}, 100]$, and $[-20, 20]$, respectively. Each panel uses four chains with 500 warm-up and 500 retained draws. The MCMC convergence diagnostic satisfies $\hat{R} \leq 1.003$ in Panel A and $\hat{R} \leq 1.004$ in Panel B, with no divergent transitions. Post. RMSD is $\{(\bar{\theta} - \theta_0)^2 + s_\theta^2\}^{1/2}$, where θ_0 , $\bar{\theta}$, and s_θ are the true value, posterior mean, and posterior standard deviation. It is conditional on the realized synthetic sample rather than a frequentist repeated-sampling quantity. The 95% credible interval contains the central 95% of posterior mass.

4.1. Parameter Recovery Under Richer Data

We examine recovery with more data in two ways: pooling independent contests generated under common parameters and simulating separate single-contest realizations with longer horizons. Both exercises retain the baseline likelihood and priors and provide finite-sample recovery diagnostics rather than formal consistency results.

Replications. We simulate independent contests under the baseline parameters and estimate their common primitives from nested pools of 10 and 20 contests. Each contest draws its own initial gap, state shocks, signal noise, and submission events. In this simulated sample, every posterior standard deviation is lower in the 20-contest pool than in the 10-contest pool. The decline for μ_0 reflects the additional independent initial-gap realizations. Post. RMSD also decreases for every parameter except c_j , for which it changes only from 0.133 to 0.134. These comparisons describe the realized nested pools and do not imply that either measure must decrease whenever the sample is enlarged.

Table 2 Bayesian Estimates from Pooled Synthetic Data

Parameters	c_i	c_j	σ	λ	μ_0
(true val.)	1.2	1.5	2.0	1.0	0.0
Pool of 10 Contests					
Posterior Mean	1.186	1.416	1.996	0.900	-0.156
Posterior SD	0.077	0.103	0.075	0.121	0.692
Post. RMSD	0.079	0.133	0.075	0.157	0.709
Pool of 20 Contests					
Posterior Mean	1.210	1.388	2.017	1.043	0.081
Posterior SD	0.057	0.072	0.052	0.105	0.558
Post. RMSD	0.058	0.134	0.055	0.113	0.564

Note: The pool of 10 contests contains 233 and 195 submissions by players i and j ; the pool of 20 contains 460 and 400. The 10-contest pool is the first half of the 20-contest pool. The priors and fixed value of r are the same as in Panel A of Table 1. Across both specifications, four chains with 500 warm-up and 500 retained draws yield an MCMC convergence diagnostic of $\hat{R} \leq 1.001$ and no divergent transitions. Post. RMSD is defined as $\{(\theta - \theta_0)^2 + s_\theta^2\}^{1/2}$ and is computed from the posterior distribution conditional on the realized synthetic sample.

Longer Contests. We next consider single simulated contests with 6- and 12-month horizons. We hold $(\theta, c_i, c_j, \sigma, \lambda, \mu_0)$ at their baseline values and calibrate the auxiliary scale r_T so that the integrated submission intensity per day along each reported path matches that of the 3-month baseline path. Because the submission intensity is $\tau_k(t) = r_T m_k(t)$, this calibration offsets the decline in average effort per day as the horizon increases. We use $r_{180} = 18.65$ and $r_{360} = 47.44$, which give a common integrated intensity of 0.618 submissions per day; realized rates remain stochastic. Table 3 shows more precise recovery of c_i , c_j , σ , and λ in the 12-month realization, whereas μ_0 remains weakly learned because each contest provides only one initial-gap realization.

Table 3 Bayesian Estimates from Longer-Horizon Synthetic Data

Parameters	c_i	c_j	σ	λ	μ_0
(true val.)	1.2	1.5	2.0	1.0	0.0
Contest of 6 Months					
Posterior Mean	1.208	1.294	2.169	1.430	-0.344
Posterior SD	0.171	0.195	0.174	0.380	0.955
Post. RMSD	0.172	0.283	0.243	0.574	1.015
Contest of 12 Months					
Posterior Mean	1.142	1.484	2.066	1.018	-0.502
Posterior SD	0.101	0.147	0.098	0.186	0.924
Post. RMSD	0.116	0.148	0.118	0.187	1.052

Note: The 6-month realization contains 59 and 55 submissions by players i and j ; the 12-month realization contains 129 and 99. The corresponding auxiliary submission scales are fixed at their calibrated values, $r_{180} = 18.65$ and $r_{360} = 47.44$. All other true parameters and the priors are the same as in Panel A of Table 1. Across both specifications, four chains with 500 warm-up and 500 retained draws yield an MCMC convergence diagnostic of $\hat{R} \leq 1.001$ and no divergent transitions. Post. RMSD is defined as $\{(\theta - \theta_0)^2 + s_\theta^2\}^{1/2}$ and is computed from the posterior distribution conditional on the realized synthetic sample.

5. Empirical Application

In this section, we estimate the posterior-aware effort model on real Kaggle contests, assess the recovered primitives against private-leaderboard-based benchmarks, and use them to evaluate counterfactual precision, prize, and duration choices.

We select representative Kaggle competitions from Meta-Kaggle and identify two focal participants i and j in each contest based on submission frequency, activity duration, and final ranking. Using their submission records and public leaderboard trajectories, we construct the observed data $\{t_k^i\}_{k=1}^{N_i}$, $\{t_k^j\}_{k=1}^{N_j}$ and $\{\hat{y}_{t_k}\}_{k=1}^{N_i+N_j}$ for contest-level Bayesian estimation under a unified prior specification. The estimation uses equation (5), so signal precision affects both the observation noise and contestants' incentives through $Q_t(\lambda)$.

The estimation likelihood uses only contest-phase submission timing and public leaderboard trajectories. Private-leaderboard path values are excluded from estimation, but final private ranks are used to select the focal pair. We therefore interpret the resulting evidence as diagnostic validation conditional on the selected pair, rather than as validation based on a fully independent sample. Specifically, we compare structural contest-level estimates of λ and σ against benchmark quantities constructed from private-leaderboard path values excluded from the likelihood. By the private-leaderboard path, we mean the ex post sequence of private scores associated with the contestants' historical submissions, reconstructed after the contest; it is not a path observed by contestants in real time. Figure 4 summarizes the data-model structure and validation design.

5.1. Preparing Data for Structural Estimation

To empirically implement the structural estimation framework developed in the previous sections using Kaggle data, the first step is to map the observed data onto the corresponding variables defined by the model. This subsection details the procedures for selecting competitions, identifying key participants, and incorporating contest-specific information such as prize structures to make the data compatible with the model's structural assumptions.

As a first selection criterion, we focus on monetary prize structure. We restrict attention to Kaggle competitions offering one, two, or three monetary prizes denominated in U.S. dollars. Competitions with a single prize naturally conform to the winner-take-all incentive structure. For competitions with two or three prize tiers, we align the empirical setting with the model by treating the prize difference between the top two contestants as the effective gain from winning. Because non-winning contestants may still receive symbolic recognition such as medals, focusing on the top-two prize gap helps mitigate confounding incentives and brings the empirical setting closer to the model's winner-take-all benchmark.

The second criterion identifies two highly active, top-ranked contestants whose participation overlaps substantially during the contest. We select candidate focal contestants beginning from

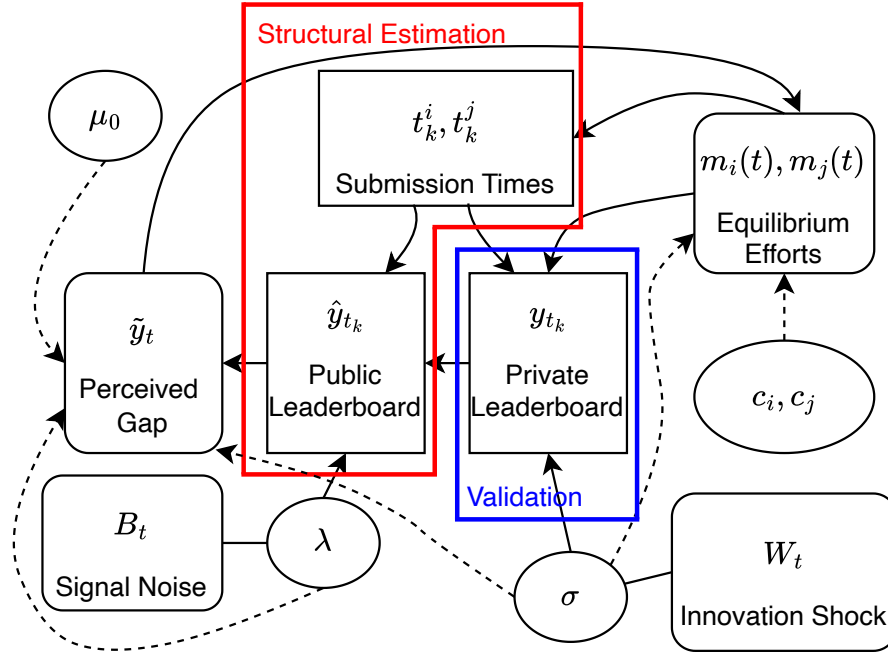


Figure 4 Meta-Kaggle Dataset and the Model Structure

Note: This figure shows the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles represent observed data, rounded rectangles represent latent states, and ellipses represent contest parameters. Solid lines depict temporal relationships among time series variables and state variables, while dashed arrows show the influence of contest parameters on the data-generating process. The private leaderboard y_t is not observed by contestants during the contest. Its path values are excluded from the likelihood and used for post-estimation validation, while final private ranks enter focal-pair selection.

the top of the final private leaderboard. This focuses the two-player model on rivalry among high-performing contestants. Private path values do not enter the estimation likelihood, but final private ranks determine the focal pair; hence, the validation exercises are path-wise holdout diagnostics conditional on ex post pair selection. To ensure overlapping activity and adequate data for estimation, each contestant must submit at least five times, and the pair must satisfy the temporal-overlap criterion used in sample construction.

While analytically convenient, restricting attention to two participants may limit representativeness, oversimplify the strategic environment, and introduce selection concerns via *ex post* rankings. We therefore interpret the two-player model as a structural approximation of focal rivalry near the top of contests with concentrated competition, frequent submissions, and overlapping activity windows.

The third criterion for contest selection concerns the format in which leaderboard scores are presented. Since Kaggle competitions are based on diverse real-world tasks, the scoring metrics vary accordingly. Some are based on raw scores (positive real numbers), while others are based on accuracy measures (expressed as percentages or decimals between 0 and 1). Ignoring these

differences would lead to inconsistencies in units across contests, complicating any meaningful cross-contest comparison of parameter estimates.

We restrict attention to contests reporting accuracy-based scores, which provides a common bounded score domain and facilitates a consistent transformation across contests. The retained focal paths are restricted to score observations satisfying the domain $p_t \in [0.5, 1)$ used below. To map raw leaderboard accuracies into the model’s state scale, we first transform each individual leaderboard score using the logit (log-odds) function:

$$x_t = a \cdot \log \left(\frac{p_t}{1 - p_t} \right), \quad p_t \in [0.5, 1) \quad (18)$$

where p_t is an individual leaderboard score, x_t is the corresponding transformed score, and a is a scaling parameter in the transformation.⁷ For the focal pair (i, j) , we then construct the model variables as transformed-score gaps: $\hat{y}_t = \hat{x}_t^i - \hat{x}_t^j$ for the public leaderboard and $y_t = x_t^i - x_t^j$ for the private leaderboard. We construct hourly evaluation grids by carrying forward each contestant’s most recently observed transformed score between submissions. Because submissions are asynchronous, only the submitting contestant’s score is updated, while the opponent’s most recent score is carried forward. The observed public gap may therefore compare scores submitted at different times and only approximates the contemporaneous latent performance gap. Our observation equation treats this gap as an approximate readout of the latent competitive state; this approximation is most credible for focal pairs with frequent submissions and substantial overlap in active periods. On the domain used here, the log-odds transformation is increasing and convex, so as p_t approaches 1, a given increase in accuracy produces a larger change in the transformed score x_t .

5.2. Private-Leaderboard Primitive Validation

We select 73 contests from the Meta-Kaggle database, encompassing all competitions concluded before April 15, 2025 that satisfy our eligibility filters on leaderboard availability, public-private score information, and focal-player activity.⁸ For each selected contest, we identify two highly active top-ranked contestants who meet our focal-pair selection criteria. Appendix C reports the Bayesian estimation results for each contest individually.

⁷ Changing a rescales the transformed score and can affect scale-dependent likelihood and validation quantities, but it does not change scale-invariant shape diagnostics of the transformed distribution. In this paper, to preserve comparability across contests and avoid extra tuning degrees of freedom, we fix $a = 1$ throughout.

⁸ The Meta-Kaggle database is updated daily, with new competition data added after the contests conclude.

Table 4 Public–Private Leaderboard Agreement

Statistic	Mean	Median	95% CI for Mean
Public–private rank correlation	0.937	0.984	[0.903, 0.963]
Top-2 overlap rate	65.1%	50.0%	[56.2%, 73.3%]
Top-5 overlap rate	66.6%	80.0%	[59.7%, 73.2%]

Note: The sample consists of the same 73 Kaggle contests used in the structural analysis. For each contest, we use the official public and private leaderboard ranks in Meta-Kaggle, excluding benchmark teams and teams missing either rank. Public–private rank correlation is the Spearman correlation across all retained teams within a contest. The top- k overlap rate is $|\mathcal{T}_k^{\text{pub}} \cap \mathcal{T}_k^{\text{pri}}|/k$, where $\mathcal{T}_k^{\text{pub}}$ and $\mathcal{T}_k^{\text{pri}}$ are the sets of teams ranked in the public and private top k , respectively; the ordering of teams within the top- k sets need not coincide. Each statistic is first calculated separately by contest, and the table reports the equally weighted mean and median across the 73 contests. The 95% confidence intervals pertain to the across-contest means and are percentile intervals obtained from 10,000 nonparametric bootstrap resamples at the contest level.

We next assess the estimated contest primitives using private-leaderboard-based benchmarks. The 146 full- and early-sample estimates contain no divergent transitions, have a maximum MCMC convergence diagnostic of $\hat{R} = 1.006$, and do not approach the common lower bound of 0.01. Across the 146 player–contest estimates, the posterior median of w_k has a median of 0.571 and is below one in 95 cases. At the stricter contest level, the posterior median of $\max\{w_i, w_j\}$ is below one in 42 of 73 contests. The leading-order approximation therefore describes most player-level estimates, but it need not be accurate for every contest; the counterfactuals below should be read as leading-order design benchmarks. Table 4 is computed directly from leaderboard ranks and therefore does not use the structural estimates.

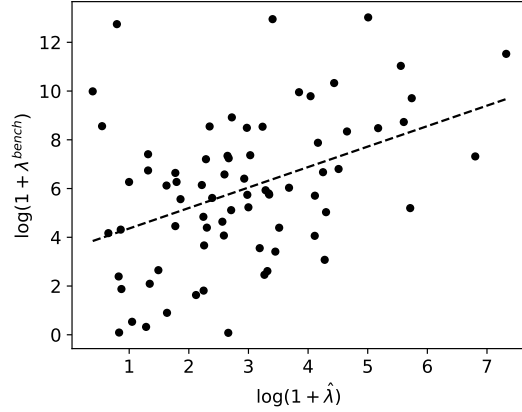
5.2.1. Private-Leaderboard Diagnostic for Signal Precision λ . The signal precision parameter λ is contest-specific and captures the informativeness of public leaderboard feedback relative to latent performance. We compare the structural estimate obtained from submissions and public leaderboard paths with an approximate benchmark constructed from public-private leaderboard discrepancies. This benchmark is informative but not fully independent of the maintained structural setup, because it still relies on the transformed leaderboard scale used in the model.

Let $\ell \in \{1, \dots, L\}$ index contests. Under the maintained Gaussian discrepancy approximation induced by the discretized signal readout, we construct the following signal-precision benchmark. Let $\mathcal{E}_\ell$ denote the ordered set of submission-triggered leaderboard readouts for the focal pair, and let $N^\ell = |\mathcal{E}_\ell|$. Let $h_k^\ell = t_k^\ell - t_{k-1}^\ell$ denote the elapsed time associated with readout t_k^ℓ . Under the maintained approximation

$$\hat{y}_{t_k}^\ell - y_{t_k}^\ell \sim \mathcal{N}\left(0, \frac{1}{\lambda_\ell h_k^\ell}\right), \quad t_k^\ell \in \mathcal{E}_\ell,$$

we construct

$$\lambda_\ell^{\text{bench}} = \frac{N^\ell}{\sum_{t_k^\ell \in \mathcal{E}_\ell} h_k^\ell (y_{t_k}^\ell - \hat{y}_{t_k}^\ell)^2}.$$



	coef.	stderr	95% CI
const.	3.521***	0.843	[1.868, 5.174]
$\log(1 + \hat{\lambda})$	0.840***	0.238	[0.374, 1.306]
Obs.		73	
R^2		0.175	
Adj. R^2		0.163	

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.
(2) OLS residual diagnostics: Omnibus = 3.913, Jarque–Bera = 3.146, Skewness = 0.482, Kurtosis = 3.325. (3) Other diagnostic statistics: Condition Number = 7.622, Durbin–Watson = 2.046.

Figure 5 $\lambda_\ell^{\text{bench}}$ vs. Bayesian Estimate $\hat{\lambda}_\ell$

Note. This figure plots the approximate contest-specific signal-precision benchmark $\lambda_\ell^{\text{bench}}$ against the Bayesian structural estimate $\hat{\lambda}_\ell$ for each of the 73 contests. The benchmark is computed from public-private leaderboard discrepancies at submission-triggered readouts as $\lambda_\ell^{\text{bench}} = N^\ell / \sum_{t_k \in \mathcal{E}_\ell} h_k^\ell (y_{t_k}^\ell - \hat{y}_{t_k}^\ell)^2$, where $N^\ell = |\mathcal{E}_\ell|$ is the number of readouts used in contest ℓ and $h_k^\ell = t_k^\ell - t_{k-1}^\ell$ is the elapsed time associated with readout t_k^ℓ . Because carried-forward scores may induce dependence across successive readouts, the benchmark should be interpreted as an approximate measure of signal precision. The dashed line is the fitted regression of $\log(1 + \lambda_\ell^{\text{bench}})$ on $\log(1 + \hat{\lambda}_\ell)$.

The benchmark therefore uses only submission-triggered leaderboard readouts rather than repeated hourly carry-forward observations.⁹ Because asynchronous updates may repeatedly use an opponent’s carried-forward score, the discrepancies can be serially dependent; we therefore interpret $\lambda_\ell^{\text{bench}}$ as an approximate signal-precision benchmark.

This benchmark relies on the Gaussian discrepancy approximation applied to the transformed scores in (18). The transformation is adopted for tractability and should be interpreted as an approximation rather than as a claim of exact normality for every contest. Auxiliary checks with alternative choices of the scale parameter a assess sensitivity to transformed-score units, not sensitivity to distributional shape. The Gaussian discrepancy structure therefore remains a maintained modeling assumption.

We then examine the cross-contest association between the private-leaderboard benchmark $\lambda_\ell^{\text{bench}}$ and the Bayesian structural estimate $\hat{\lambda}_\ell$. This exercise tests the observation-precision interpretation of λ . It does not by itself validate the behavioral channel through which λ enters $Q_t(\lambda)$ and effort.

The estimated slope is 0.840 with a 95% confidence interval of [0.374, 1.306]. The positive association also appears in ranks (Spearman $\rho = 0.377$, $p = 0.001$). Thus, contests estimated to have more precise public feedback also tend to have a larger precision benchmark. The relationship is

⁹ As a robustness comparison to the irregular-interval estimator above, we also consider the fixed-grid approximation $\lambda_{\ell, \Delta}^{\text{bench}} = N^\ell / \left\{ \Delta \sum_{t_k^\ell \in \mathcal{E}_\ell} (y_{t_k}^\ell - \hat{y}_{t_k}^\ell)^2 \right\}$. Despite this difference, both constructions yield the same qualitative conclusion of a positive but modest cross-contest association; the fixed-grid construction also underlies the alternative focal-pair validation in Section 6.1.

informative but not exact: the benchmark is approximate and the structural estimate also uses submission behavior.

5.2.2. Public-Private Leaderboard Disagreement Diagnostics. We next provide an additional diagnostic for signal precision based directly on public-private leaderboard disagreement, without first converting the disagreement into a Gaussian precision benchmark. If $\hat{\lambda}_\ell$ captures public-signal precision, contests with larger $\hat{\lambda}_\ell$ should exhibit smaller disagreement between the public leaderboard path and the corresponding private-leaderboard path. We therefore construct the following contest-level outcome:

$$D_\ell^{path} = \frac{1}{|\mathcal{H}_\ell|} \sum_{h \in \mathcal{H}_\ell} |y_{\ell h} - \hat{y}_{\ell h}|,$$

where $\mathcal{H}_\ell$ denotes the set of hourly observation points used in the empirical leaderboard path, and D_ℓ^{path} measures the mean absolute public-private gap over the contest path. The predicted association is therefore negative.

Because the full-path diagnostic uses the same contest path both to estimate $\hat{\lambda}_\ell$ and to construct D_ℓ^{path} , it may partly reflect common path-level variation rather than purely predictive content. We therefore also report a time-split version of this diagnostic. Let $\hat{\lambda}_\ell^{early}$ denote the structural estimate obtained using only the first τT_ℓ hours of contest-phase data, with $\tau = 0.75$. The validation target is then constructed only from the remaining public-private path:

$$D_\ell^{late} = \frac{1}{|\mathcal{H}_\ell^{late}|} \sum_{h \in \mathcal{H}_\ell^{late}} |y_{\ell h} - \hat{y}_{\ell h}|.$$

Here $\mathcal{H}_\ell^{late} = \{h \in \mathcal{H}_\ell : h > \tau T_\ell\}$. For both diagnostics, we estimate the log-log validation slope linking the discrepancy measure to the corresponding signal-precision estimate. The full-path specification uses $\log(1 + D_\ell^{path})$ and $\log(1 + \hat{\lambda}_\ell)$, while the time-split specification uses $\log(1 + D_\ell^{late})$ and $\log(1 + \hat{\lambda}_\ell^{early})$. The latter asks whether signal precision estimated from early public contest information predicts public-private disagreement measured over the later contest period, conditional on the focal-pair selection rule.

Both estimates have the predicted negative sign. The full-path association is stronger, but the early estimate also predicts lower public-private disagreement in the held-out final quarter. The time-split coefficient is less precise and explains less cross-contest variation, as expected from using shorter estimation and validation windows.

Table 5 Signal Precision and Public-Private Path Disagreement

Specification	coef.	stderr	95% CI	R ²	Spearman ρ
Full path	-0.040**	0.013	[-0.066, -0.013]	0.111	-0.341**
Time split	-0.037*	0.016	[-0.068, -0.007]	0.030	-0.258*
Obs.	73				

Note: For contest ℓ , $D_\ell^{path} = |\mathcal{H}_\ell|^{-1} \sum_{h \in \mathcal{H}_\ell} |y_{\ell h} - \hat{y}_{\ell h}|$ measures the mean absolute gap between private and public leaderboard paths. The full-path row reports the slope from $\log(1 + D_\ell^{path}) = \alpha + \beta \log(1 + \hat{\lambda}_\ell) + \varepsilon_\ell$, where $\hat{\lambda}_\ell$ is the baseline structural estimate. For the time-split row, $\hat{\lambda}_\ell^{early}$ is estimated using only the first 75% of contest-phase data, and $D_\ell^{late} = |\mathcal{H}_\ell^{late}|^{-1} \sum_{h \in \mathcal{H}_\ell^{late}} |y_{\ell h} - \hat{y}_{\ell h}|$ is measured using only the remaining 25% of the private and public leaderboard paths. The reported coefficient is the slope from $\log(1 + D_\ell^{late}) = \alpha + \beta \log(1 + \hat{\lambda}_\ell^{early}) + \varepsilon_\ell$. (1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (2) OLS residual diagnostics (Full path; Time split): Omnibus = 47.496; 77.233, Jarque–Bera = 130.326; 563.537, Skewness = 2.223; 3.324, Kurtosis = 7.804; 14.877. (3) Other diagnostic statistics (Full path; Time split): Condition Number = 7.622; 6.811, Durbin–Watson = 2.118; 2.116.

5.2.3. Private-Leaderboard Consistency Benchmark for Innovation Risk σ . Another key contest-specific parameter is σ , which captures innovation uncertainty and the extent to which contest outcomes are driven by chance rather than effort. For each contest ℓ , we estimate $\hat{\sigma}_\ell$ structurally using the two contestants’ public submission records $\{t_k^\ell\}_{k=1}^{N_i^\ell + N_j^\ell}$ and public leaderboard trajectories $\{\hat{y}_k^\ell\}_{k=1}^{N_i^\ell + N_j^\ell}$. We then construct a private-leaderboard-based benchmark for σ using score increments.

Let $(t_k^\ell, y_k^\ell)_{k=0}^{n_\ell}$ denote the private-leaderboard gap evaluated at the ordered update times used for this benchmark, including the normalized initial observation. Under the diffusion specification in (1), the private-leaderboard increment satisfies $y_k^\ell - y_{k-1}^\ell \mid \{m_i^\ell, m_j^\ell\} \sim \mathcal{N}(\mu_k^\ell, \sigma^2(t_k^\ell - t_{k-1}^\ell))$ for $k = 1, \dots, n_\ell$. Here, $\mu_k^\ell = \int_{t_{k-1}^\ell}^{t_k^\ell} [m_i^\ell(s) - m_j^\ell(s)] ds$. Motivated by the Gaussian increment formula, we compute the following contest-specific consistency benchmark; the corresponding diffusion calculation is provided in Appendix B.2:

$$\sigma_\ell^{\text{bench}} = \left(\frac{1}{n_\ell} \sum_{k=1}^{n_\ell} \frac{[y_k^\ell - y_{k-1}^\ell - \hat{\mu}_k^\ell]^2}{t_k^\ell - t_{k-1}^\ell} \right)^{1/2}$$

The drift adjustment $\hat{\mu}_k^\ell$ is computed from the Bayesian posterior effort trajectories, so this benchmark is model-assisted rather than independent. In addition, because contestant submissions are asynchronous, consecutive displayed private-gap observations may combine a new score for one contestant with a carried-forward score for the other. The resulting estimator should therefore be interpreted as an approximate consistency benchmark, not as an exact maximum likelihood estimator of latent diffusion volatility. For this reason, we place greater weight on the less model-dependent private-path volatility diagnostics reported below.

As in the analysis above for λ , we regress the benchmark σ^{bench} on the structural estimate $\hat{\sigma}$ to examine whether a positive relationship exists between the two.

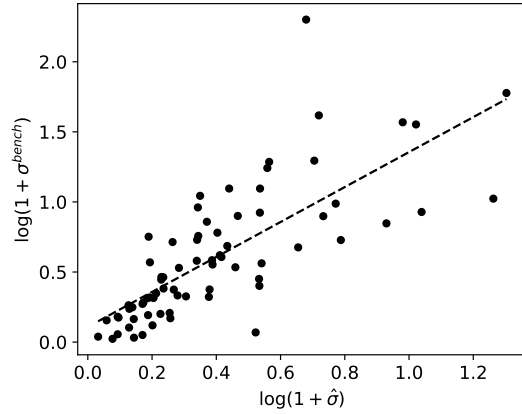


Figure 6 σ^{bench} vs. Bayesian Estimate $\hat{\sigma}$

Note. This figure plots the private-leaderboard consistency benchmark σ^{bench} against the Bayesian structural estimate $\hat{\sigma}$ for each of the 73 contests. The benchmark is computed from carried-forward private leaderboard increments as $\sigma^{\text{bench}} = \left(\frac{1}{n} \sum_{k=1}^n \frac{(y_k - y_{k-1} - \hat{\mu}_k)^2}{t_k - t_{k-1}} \right)^{1/2}$, where n is the number of adjacent displayed-gap increments and $\hat{\mu}_k$ is computed from posterior effort trajectories. Because asynchronous observations can combine a new score with an opponent's stale score, the quantity is an approximate, model-assisted consistency benchmark rather than an exact diffusion MLE. The dashed line represents the fitted regression line from $\log(1 + \sigma^{\text{bench}}) = \beta_0 + \beta_1 \log(1 + \hat{\sigma}) + \varepsilon$.

The estimated slope is 1.247 with a 95% confidence interval of [0.938, 1.557]. The rank association is also strong (Spearman $\rho = 0.827$, $p < 0.001$). Contests with larger estimated innovation risk therefore exhibit larger drift-adjusted private-leaderboard innovations.

5.2.4. Private-Leaderboard Volatility Diagnostics. The preceding validation of σ uses an approximate private-leaderboard consistency benchmark motivated by the normal increment structure of the diffusion model. That benchmark is parametric and uses the posterior drift adjustment $\hat{\mu}_k$. We therefore add a simpler diagnostic based on realized private-leaderboard volatility. For each contest ℓ , let $\Delta y_{\ell h} = y_{\ell h} - y_{\ell, h-1}$ denote the increment on the constructed hourly evaluation grid, and define $V_{\ell}^{\text{path}} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_{\ell}, h > 1\}$. This statistic does not require estimating a normal-increment likelihood or subtracting a model-implied drift term. It asks whether contests with larger estimated innovation risk also exhibit more volatile private-leaderboard paths. Because the hourly private paths are constructed by carrying scores forward between submissions, the resulting series contains zero increments between updates and discrete jumps at submission times. This volatility measure therefore reflects both changes in submitted scores and the timing of private-leaderboard updates. We interpret it as a less model-dependent path diagnostic, not as a direct measure of latent diffusion volatility. We estimate this relationship in logs, using $\log(1 + V_{\ell}^{\text{path}})$ and $\log(1 + \hat{\sigma}_{\ell})$, with the predicted positive coefficient.

We also report a time-split version of the same diagnostic. Let $\hat{\sigma}_{\ell}^{\text{early}}$ denote the structural estimate obtained using only the first τT_{ℓ} hours of contest-phase data, with $\tau = 0.75$. The validation target is then constructed only from the remaining private-leaderboard path, $V_{\ell}^{\text{late}} = \text{sd}\{\Delta y_{\ell h} : h \in$

	coef.	stderr	95% CI
const.	0.108*	0.049	[0.011, 0.205]
$\log(1 + \hat{\sigma})$	1.247***	0.158	[0.938, 1.557]
Obs.		73	
R^2		0.574	
Adj. R^2		0.568	

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

(2) OLS residual diagnostics: Omnibus = 27.225, Jarque-Bera = 70.484, Skewness = 1.149, Kurtosis = 7.229. (3) Other diagnostic statistics: Condition Number = 4.180, Durbin-Watson = 1.883.

$\mathcal{H}_\ell$, $h > \tau T_\ell$. For the time-split specification, we use $\log(1 + V_\ell^{late})$ and $\log(1 + \hat{\sigma}_\ell^{early})$. Conditional on the focal-pair selection rule, the time-split version provides a stricter temporal separation because the structural estimate uses only early public contest data, while the validation outcome is measured on the later private-leaderboard path.

Table 6 Private-Leaderboard Volatility Diagnostics

Specification	coef.	stderr	95% CI	R ²	Spearman ρ
Full path	0.327***	0.057	[0.215, 0.439]	0.611	0.845***
Time split	0.201*	0.090	[0.025, 0.377]	0.127	0.316**
Obs.	73				

Note: For contest ℓ , $y_{\ell h}$ denotes the private-leaderboard gap on the constructed hourly evaluation grid, and $\Delta y_{\ell h} = y_{\ell h} - y_{\ell, h-1}$ is the corresponding grid increment. The full-path row reports the slope β from $\log(1 + V_\ell^{path}) = \alpha + \beta \log(1 + \hat{\sigma}_\ell) + \varepsilon_\ell$, where $V_\ell^{path} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_\ell, h > 1\}$ and $\hat{\sigma}_\ell$ is the baseline structural estimate. The time-split row reports the slope β from $\log(1 + V_\ell^{late}) = \alpha + \beta \log(1 + \hat{\sigma}_\ell^{early}) + \varepsilon_\ell$, where $\hat{\sigma}_\ell^{early}$ is estimated using only the first 75% of contest-phase data and $V_\ell^{late} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_\ell, h > 0.75T_\ell\}$. Spearman ρ is the rank correlation between the same dependent variable and structural estimate used in each row. (1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (2) OLS residual diagnostics (Full path; Time split): Omnibus = 17.761; 32.525, Jarque–Bera = 101.881; 59.347, Skewness = -0.090; 1.672, Kurtosis = 8.785; 5.885. (3) Other diagnostic statistics (Full path; Time split): Condition Number = 4.180; 4.525, Durbin–Watson = 1.608; 1.751.

Both volatility regressions have the predicted positive sign. The full-path relationship is strong, while the early estimate retains a smaller positive association with volatility in the held-out final quarter. Because $\hat{\sigma}_\ell$ and private-path volatility are expressed in transformed-score units, these results are interpreted as consistency diagnostics rather than scale-free validation evidence.

5.3. Implication for Precision Design

We use the estimated primitives to ask whether the observed public signal is too precise for incentives and how precision design changes the prize–duration trade-off. As in Proposition ??, these leading-order counterfactuals hold the initial true-output environment fixed and impose the calibrated mapping $P^{\text{cal}}(\lambda) = \bar{S}(\lambda)$ when precision changes. The score–quality wedge is not separately estimated, so the exercise is conditional on this design-layer calibration. The first exercise also holds the observed prize and duration fixed. For posterior draw s in contest ℓ , Proposition ?? gives the precision best response

$$\lambda_{\ell s}^{\text{BR}}(T) = \begin{cases} \infty, & |\mu_{0, \ell s}| \leq \sigma_{\ell s} \sqrt{T}, \\ \left[\frac{\sigma_{\ell s}}{\mu_{0, \ell s}^2 - \sigma_{\ell s}^2 T} \right]^2, & |\mu_{0, \ell s}| > \sigma_{\ell s} \sqrt{T}. \end{cases}$$

We summarize each contest by the posterior probability that finite precision is optimal. This probability is more informative than a plug-in classification because the threshold depends on the initial public gap, which can be imprecisely estimated from a single contest path.

The second exercise compares prize–duration frontiers with fixed and reoptimized precision. Let $B_{0,\ell s} = \tilde{B}(\mu_{0,\ell s}, \theta_\ell^{obs}, T_\ell^{obs}, \lambda_{\ell s})$ denote predicted best true output under the observed design. For $T_{a,\ell} = aT_\ell^{obs}$, define

$$\theta_{a,\ell s}^{BO,p} = \min \{ \theta \in [\theta_{\min}, \theta_\ell^{obs}] : \tilde{B}(\mu_{0,\ell s}, \theta, T_{a,\ell}, \lambda_{a,\ell s}^p) \geq B_{0,\ell s} \},$$

where $p \in \{\text{fixed}, \text{optimized}\}$, $\lambda_{a,\ell s}^{\text{fixed}} = \lambda_{\ell s}$, and $\lambda_{a,\ell s}^{\text{optimized}} = \lambda_{\ell s}^{\text{BR}}(T_{a,\ell})$. We set $\theta_{\min} = 0.0001$ in hundred-thousand-dollar units, equivalent to ten U.S. dollars. Comparing the two frontiers isolates the additional prize reduction made possible by redesigning public feedback. The exercise holds the duration change fixed and does not claim that longer contests are costless or welfare improving.

The closed-form objective in Lemma 2 assumes symmetric costs. In the main counterfactual, we project the estimated pair onto that benchmark using the inverse-cost harmonic mean $c_{\ell s}^{sym} = 2/(c_{i,\ell s}^{-1} + c_{j,\ell s}^{-1})$, which preserves the leading-order coefficient on total effort. We treat this as a transparent design benchmark rather than an exact asymmetric extension. An asymmetric posterior simulation is reserved for robustness analysis.

Table 7 Posterior Precision Recommendations

Posterior criterion	Any finite	> 50%	> 75%	> 90%
Finite-precision probability	55	6	2	0

Note: Entries report the number of contests whose posterior probability of $|\mu_0| > \sigma\sqrt{T^{obs}}$ is positive or exceeds the indicated threshold.

Table 8 Best-Output-Preserving Prize–Duration–Precision Substitution

Duration	Precision rule	Lower prize	$\geq 5\%$ lower	Mean change	Median change
$1.00T^{obs}$	Fixed	0	0	0.00%	0.00%
$1.00T^{obs}$	Optimized	73	2	-1.37%	-1.04%
$1.01T^{obs}$	Fixed	73	14	-3.54%	-1.71%
$1.01T^{obs}$	Optimized	73	19	-4.84%	-2.90%
$1.05T^{obs}$	Fixed	73	55	-15.27%	-8.27%
$1.05T^{obs}$	Optimized	73	60	-16.32%	-9.07%
$1.10T^{obs}$	Fixed	73	73	-25.36%	-15.95%
$1.10T^{obs}$	Optimized	73	73	-26.19%	-16.73%

Note: For each posterior draw, the fixed rule holds precision at its estimated observed value, whereas the optimized rule uses the precision best response at the counterfactual duration. Prize changes are measured relative to θ^{obs} . “Lower prize” and “ $\geq 5\%$ lower” report numbers of contests; changes first average across posterior draws within contest and then across contests.

Reoptimizing precision reduces the required prize even at the observed duration, although the median reduction is only 1.04%. Extending duration has a larger effect: a 5% extension lowers the median required prize by 8.27% under fixed precision and 9.07% when precision is reoptimized. Precision design therefore adds to, but does not replace, the prize reduction generated by a longer contest.

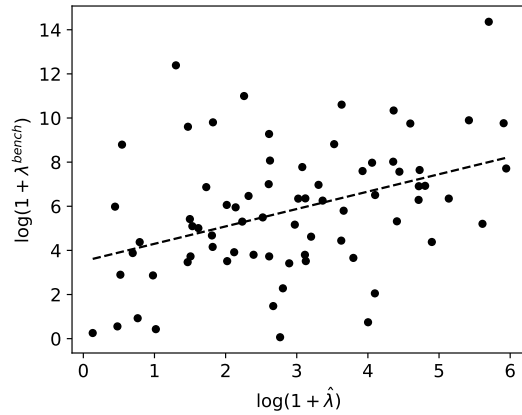
6. Robustness

This section examines whether the validation results depend on the selected focal pair. We re-estimate the model using a lower-ranked eligible pair and compare the resulting validation patterns with the baseline findings in Section 5.2. We then clarify the separate limitation imposed by the maintained Gaussian structure.

6.1. Selection of Two Contestants

The baseline analysis pairs two highly active top-ranked contestants in each contest. The robustness sample instead uses the second and third eligible contestants in final private-leaderboard order, subject to the same minimum-activity and overlap requirements. This shifts the focal rivalry away from the leading eligible contestant while preserving the 73-contest sample.

In 56 contests, the selected pair is exactly the second- and third-ranked contestants. When either contestant does not meet the eligibility requirements, we move down the final private leaderboard to the next eligible contestant.



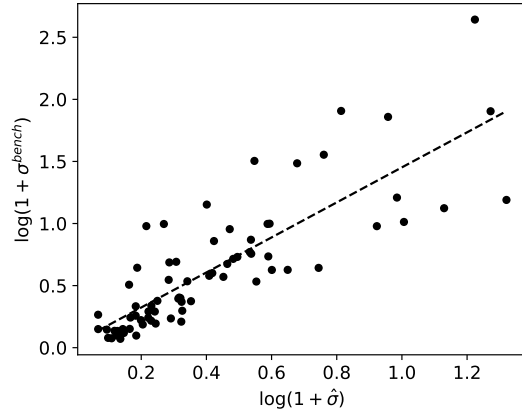
	coef.	stderr	95% CI
const.	3.510***	0.771	[1.999, 5.022]
$\log(1 + \hat{\lambda})$	0.788***	0.230	[0.337, 1.240]
Obs.		73	
R^2		0.157	
Adj. R^2		0.145	

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.
(2) OLS residual diagnostics: Omnibus = 3.494, Jarque–Bera = 2.700, Skewness = 0.432, Kurtosis = 3.375. (3) Other diagnostic statistics: Condition Number = 7.729, Durbin–Watson = 2.200.

Figure 7 Robustness Check (2nd–3rd Eligible Pair): $\lambda_\ell^{\text{bench}}$ vs. Bayesian Estimate $\hat{\lambda}_\ell$

Note. This figure repeats the signal-precision validation using the second and third eligible contestants in final private-leaderboard order. Eligibility requires at least five submissions and at least ten days of overlapping activity. The selected pair is exactly the second and third final ranks in 56 of the 73 contests; otherwise, the next-ranked eligible contestant is used. The dashed line is the fitted regression $\log(1 + \lambda_\ell^{\text{bench}}) = \beta_0 + \beta_1 \log(1 + \hat{\lambda}_\ell) + \varepsilon_\ell$.

The signal-precision result is nearly unchanged under the alternative pairing rule. The slope is 0.788 ($p < 0.001$), compared with 0.840 in the baseline, and the corresponding R^2 values are 0.157 and 0.175. The positive but moderate relationship therefore does not depend on including the leading eligible contestant.



	coef.	stderr	95% CI
const.	0.041	0.061	[-0.08, 0.16]
$\log(1 + \hat{\sigma})$	1.411***	0.183	[1.053, 1.769]
Obs.	73		
R^2	0.687		
Adj. R^2	0.682		

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

(2) OLS residual diagnostics: Omnibus = 11.712, Jarque-Bera = 12.344, Skewness = 0.839, Kurtosis = 4.114. (3) Other diagnostic statistics: Condition Number = 3.929, Durbin-Watson = 1.753.

Figure 8 Robustness Check (2nd–3rd Eligible Pair): σ^{bench} vs. Bayesian Estimate $\hat{\sigma}$

Note. This figure repeats the innovation-risk validation using the same alternative focal-pair rule as Figure 7. The dashed line is the fitted regression $\log(1 + \sigma^{\text{bench}}) = \beta_0 + \beta_1 \log(1 + \hat{\sigma}) + \varepsilon$.

The innovation-risk relationship also remains strong. The alternative-pair slope is 1.411 ($p < 0.001$), compared with 1.247 in the baseline, while R^2 rises from 0.574 to 0.687. Together, the two exercises show that the validation patterns are not driven by the leading eligible contestant.

6.2. Normality Caveat and Interpretation

The multiplicative parameter a in (18) rescales transformed scores and can affect scale-dependent likelihood and validation quantities, but it cannot alter scale-invariant normality diagnostics. Varying a therefore provides a scale-sensitivity check, not a test of the Gaussian approximation.

We do not report a non-Gaussian re-estimation because the filtering equation, likelihood, and benchmark constructions all rely on the maintained Gaussian structure. A meaningful distributional robustness exercise would require replacing these components jointly rather than changing the score scale alone.

The focal-pair exercise supports the use of a two-player representation for leading rivalries. The Gaussian structure remains a maintained modeling restriction, so the private-leaderboard comparisons should still be interpreted as validation within that structure.

7. Conclusion

This paper develops a dynamic game-theoretic model and Bayesian structural estimation framework for online crowdsourcing contests under noisy public feedback, with the final private leaderboard determining the realized winner. Contestants evaluate the prize using their full posterior winning probabilities, yielding a closed-form leading-order equilibrium in which signal precision determines how broadly incentives extend across public performance gaps. Conditional on focal pairs selected using final private ranks, we recover latent competition states, effort dynamics, and

contest-specific primitives from contest-phase submission timing and public leaderboard trajectories, and assess the estimated primitives using private-leaderboard path values excluded from the likelihood. Under the calibrated score–quality mapping that holds the initial true-output environment fixed, the leading-order design analysis shows that perfect precision is optimal in a close contest, whereas a finite precision sustains effort when the initial public gap exceeds the innovation risk accumulated before the deadline.

Several limitations deserve mention. First, the posterior-aware equilibrium is a leading-order characterization for small dimensionless incentive indices $w_k = \theta/(\sigma^2 c_k)$; a complete nonlinear solution remains an open extension. Second, the empirical setting differs from the baseline model in several respects: many contests use multi-tier or team-based prizes, actual contests involve more than two strategic contestants, leaderboard metrics vary across contests, and the focal pair is selected using final private ranks. Third, the empirical model approximates asynchronous submission-triggered updates with a continuous public signal and imposes the calibrated initialization $S_0 = \bar{S}(\lambda)$. Finally, the design result conditions on the common initial assessment, holds the initial true-output environment fixed across precision counterfactuals, and relies on the score–quality calibration in equation (??). These limitations suggest natural extensions involving nonlinear equilibrium computation, richer prize structures, nonstationary filtering, and multi-agent dynamics.

Appendix A: Proofs

Proof of Theorem 1 Under the calibrated initialization, the innovation representation is

$$d\tilde{y}_t = (q_{i,t} - q_{j,t})dt + \sigma d\hat{B}_t.$$

Thus, the controlled state, quadratic costs, and HJB system have the same form as in Ryvkin (2022). The difference is the terminal payoff: Ryvkin’s winner indicator is replaced here by the posterior winning probabilities G_i and G_j . Write

$$V^k = \theta(v^k + R^k), \quad k \in \{i, j\},$$

where the regular-expansion condition in Theorem 1 gives $R_t^k, R_y^k, R_{yy}^k = O(\bar{w})$ uniformly. Substitution into the HJB system shows that, relative to the diffusion term, each quadratic strategic term is multiplied by an incentive index $w_k = \theta/(\sigma^2 c_k)$ and is therefore $O(\bar{w})$. The leading normalized values solve

$$v_t^k + \frac{\sigma^2}{2} v_{yy}^k = 0.$$

Their terminal conditions are

$$v^i(y, T) = \Phi\left(\frac{y}{\sqrt{\bar{S}}}\right), \quad v^j(y, T) = \Phi\left(-\frac{y}{\sqrt{\bar{S}}}\right).$$

Let Z be standard normal and put $\tau = T - t$. The heat-semigroup representation and the Gaussian convolution identity give

$$v^i(y, t) = \mathbb{E}\left[\Phi\left(\frac{y + \sigma\sqrt{\tau}Z}{\sqrt{\bar{S}}}\right)\right] = \Phi\left(\frac{y}{\sqrt{\bar{S} + \sigma^2\tau}}\right),$$

with $v^j(y, t) = v^i(-y, t)$. Hence

$$v_y^i(y, t) = -v_y^j(y, t) = \frac{1}{\sqrt{2\pi Q_t}} \exp\left(-\frac{y^2}{2Q_t}\right), \quad Q_t = \bar{S} + \sigma^2(T - t).$$

This is the only change to Ryvkin's leading kernel: the residual variance $\sigma^2(T - t)$ is replaced by $Q_t = \sigma^2(T - t) + \bar{S}$. Finally, the same effort first-order conditions give

$$m_i = \sigma^2 w_i [v_y^i + O(\bar{w})], \quad m_j = \sigma^2 w_j [-v_y^j + O(\bar{w})].$$

Because $w_k \leq \bar{w}$, substitution of the displayed derivatives yields the $O(\bar{w}^2)$ remainder in equation (5). For fixed finite λ , $Q_t \geq \bar{S} > 0$; the leading derivative is bounded through the terminal limit, while uniformity of the remainder follows from the regular-expansion condition. $\square$

Proof of Lemma 1 Under the calibrated initialization, the order-zero innovation representation is

$$d\tilde{y}_s = \sigma d\hat{B}_s.$$

Thus, conditional on $\tilde{y}_t$, $\tilde{y}_s \sim \mathcal{N}(\tilde{y}_t, \sigma^2(s - t))$ for $s \geq t$. The effort-induced drift changes the state law by $O(w)$, while equation (5) is itself of order w . Replacing the state law by its zero-order counterpart therefore changes expected effort only at order w^2 .

For $X \sim \mathcal{N}(\mu, V)$ and $q > 0$, completing the square yields

$$\mathbb{E}\left[\frac{1}{\sqrt{2\pi q}} e^{-X^2/(2q)}\right] = \frac{1}{\sqrt{2\pi(q+V)}} e^{-\mu^2/[2(q+V)]}. \quad (19)$$

Apply equation (19) with $q = \bar{S} + \sigma^2(T - s)$ and $V = \sigma^2(s - t)$. Their sum is $Q_t = \bar{S} + \sigma^2(T - t)$, independent of s . Summing the two effort expressions in equation (5), imposing $c_i = c_j = c$, and integrating over $s \in [t, T]$ gives equation (8). $\square$

Proof of Lemma 2 For compactness within the proof, define the marginal prize contribution

$$A(\tilde{y}_0, T, \lambda) := \frac{T}{c\sqrt{2\pi Q_0(\lambda)}} \exp\left[-\frac{\tilde{y}_0^2}{2Q_0(\lambda)}\right].$$

Equation (??) implies the pathwise identity

$$\max\{o_T^i, o_T^j\} = \frac{o_T^i + o_T^j}{2} + \frac{|o_T^i - o_T^j|}{2}.$$

By equation (??), the initial average true output is zero. Because the Brownian terms have conditional mean zero, Lemma 1 at $(\tilde{y}_0, 0)$ therefore gives

$$\mathbb{E}\left[\frac{o_T^i + o_T^j}{2} \mid I_0\right] = \theta A(\tilde{y}_0, T, \lambda) + O\left(\frac{w^2 \sigma^2 T}{\sqrt{Q_0(\lambda)}}\right).$$

Under symmetric costs, the leading terms of m_i and m_j coincide. Define

$$R_T := \int_0^T [m_i(\tilde{y}_s, s) - m_j(\tilde{y}_s, s)] ds.$$

The true-output gap dynamics and equation (??) imply

$$o_T^i - o_T^j = \tilde{y}_0 + \sigma W_T + R_T, \quad W_T \sim \mathcal{N}(0, T),$$

where W_T is independent of I_0 . The uniform remainder in equation (5) gives

$$\mathbb{E}[|R_T| \mid I_0] = O\left(\frac{w^2 \sigma^2 T}{\sqrt{Q_0(\lambda)}}\right).$$

The inequality $||a+b| - |a|| \leq |b|$ shows that R_T changes the expected absolute gap by no more than the stated remainder. The folded-normal formula gives

$$\frac{1}{2} \mathbb{E}[|\tilde{y}_0 + \sigma W_T| \mid I_0] = \frac{\sigma \sqrt{T}}{\sqrt{2\pi}} e^{-\tilde{y}_0^2/(2\sigma^2 T)} + |\tilde{y}_0| \left[\Phi\left(\frac{|\tilde{y}_0|}{\sigma \sqrt{T}}\right) - \frac{1}{2} \right] = \bar{\mathcal{B}}(\tilde{y}_0, 0, T, \lambda).$$

Combining the average-output and absolute-gap terms proves equation (9). $\square$

Proof of Proposition ?? For fixed $(\tilde{y}_0, T, \lambda)$, write

$$A := \bar{\mathcal{B}}_\theta(\tilde{y}_0, \theta, T, \lambda) > 0, \quad \mathcal{B}^0 := \bar{\mathcal{B}}(\tilde{y}_0, 0, T, \lambda).$$

Equation (9) shows that A is independent of θ and that $\bar{\mathcal{B}} = \mathcal{B}^0 + \theta A$. Hence the objective in equation (??) is strictly concave in θ , with derivative

$$Au'(\mathcal{B}^0 + \theta A) - 1.$$

The derivative is strictly decreasing because $A > 0$ and $u'' < 0$. If its value at zero is nonpositive, the unique optimum is $\theta^* = 0$; otherwise, the optimum is the unique positive root. Combining these two cases gives the positive-part expression in equation (??).

Suppose next that $\theta^* > 0$. Holding (θ^*, T^*) fixed, u is strictly increasing and the no-effort component $\mathcal{B}^0$ is independent of λ . Optimal precision therefore maximizes

$$A(T^*, \lambda) = \frac{T^*}{c \sqrt{2\pi Q_0(\lambda)}} \exp\left[-\frac{\tilde{y}_0^2}{2Q_0(\lambda)}\right].$$

The displayed leading term extends continuously to $\lambda = \infty$. Since $Q_0(\lambda) = \sigma^2 T^* + \sigma/\sqrt{\lambda}$, varying $\lambda \in (0, \infty]$ is equivalent to varying $Q_0 \in [\sigma^2 T^*, \infty)$. Differentiating with respect to Q_0 , while holding $(\tilde{y}_0, T^*)$ fixed, gives

$$\frac{\partial A}{\partial Q_0} = \frac{A}{2Q_0^2} (\tilde{y}_0^2 - Q_0).$$

The derivative is nonpositive throughout the feasible range when $\tilde{y}_0^2 \leq \sigma^2 T^*$, so the maximum occurs at $Q_0 = \sigma^2 T^*$, corresponding to $\lambda^* = \infty$. When $\tilde{y}_0^2 > \sigma^2 T^*$, the derivative is positive below $\tilde{y}_0^2$ and negative above it. The unique maximizer therefore satisfies $Q_0 = \tilde{y}_0^2$, or

$$\lambda^* = \left[\frac{\sigma}{\tilde{y}_0^2 - \sigma^2 T^*} \right]^2.$$

This proves the precision expression in equation (??).

At an optimum with $\theta^* > 0$, the first-order conditions for prize and duration are

$$u'(\bar{\mathcal{B}}) \bar{\mathcal{B}}_\theta = 1, \quad u'(\bar{\mathcal{B}}) \bar{\mathcal{B}}_T = D'(T^*),$$

where all derivatives are evaluated at $(\tilde{y}_0, \theta^*, T^*, \lambda^*)$. Dividing the second condition by the first gives the duration expression in equation (??). If $\theta^* = 0$, equation (9) is independent of λ , and direct differentiation with respect to T gives the stated duration condition. $\square$

A.1. Local Identification

Proof of Proposition 1 Partition the parameter vector as $\eta = (\sigma, \lambda)$ and $\vartheta = (c_i, c_j, \mu_0, r)$. Suppose that $\Theta = (\eta, \vartheta)$ and $\Theta' = (\eta', \vartheta')$ lie in a neighborhood of Θ_0 and induce the same joint likelihood for every admissible observed history.

Equality of the point-process laws first identifies the two conditional intensity functions, by uniqueness of the compensator. Hence $\tau_i(t)$ and $\tau_j(t)$ are the same under Θ and Θ' for every admissible public history. Define their cumulative difference at update time t_k by

$$A_k := \int_0^{t_k} [\tau_i(s) - \tau_j(s)] ds. \quad (20)$$

Because $m_i - m_j = (\tau_i - \tau_j)/r$, the predictable location of the k th leaderboard readout is

$$d_k(\mu_0, r) := \mu_0 + \frac{A_k}{r}. \quad (21)$$

Conditional on the submission history, the structural residual $e_k(\Theta) := \hat{y}_{t_k} - d_k(\mu_0, r)$ is Gaussian. Its covariance matrix is

$$\Sigma(\eta) = \bar{S}(\lambda) \mathbf{1}\mathbf{1}^\top + \sigma^2 K + D_\lambda, \quad K_{k\ell} = \min\{t_k, t_\ell\}, \quad (D_\lambda)_{kk} = \frac{1}{\lambda h_k}. \quad (22)$$

We next show that equality of the leaderboard likelihoods identifies both the residual covariance and the predictable locations, without presuming that either is already known. Let

$$\beta_k(\eta) := \Sigma_{k, < k}(\eta) \Sigma_{< k, < k}(\eta)^{-1}, \quad v_k(\eta) := \Sigma_{kk}(\eta) - \Sigma_{k, < k}(\eta) \Sigma_{< k, < k}(\eta)^{-1} \Sigma_{< k, k}(\eta),$$

with β_1 empty and $v_1 = \Sigma_{11}$. The Gaussian conditioning formula gives

$$\hat{y}_{t_k} | \mathcal{F}_{k-1}, \Theta \sim \mathcal{N}(d_k(\mu_0, r) + \beta_k(\eta)(\hat{\mathbf{y}}_{< k} - \mathbf{d}_{< k}(\mu_0, r)), v_k(\eta)). \quad (23)$$

Here, $\hat{\mathbf{y}}_{< k}$ and $\mathbf{d}_{< k}$ collect the first $k-1$ readouts and predictable locations, respectively. Equality of the conditional densities under Θ and Θ' implies $v_k(\eta) = v_k(\eta')$ for every k . It also implies equality of the conditional-mean functions in equation (23). On the finite horizon, the leading-order effort kernel is bounded because $Q_t(\lambda) \geq \bar{S}(\lambda) > 0$. The intensities and the cumulative terms A_k are therefore bounded over a neighborhood of the interior parameter vector. Since the public readouts have full support on $\mathbb{R}$, any nonzero difference between $\beta_k(\eta)$ and $\beta_k(\eta')$ would generate an unbounded difference between the two conditional means as $\hat{\mathbf{y}}_{< k}$ varies, whereas all remaining terms are bounded. Thus

$$\beta_k(\eta) = \beta_k(\eta') \quad \text{and} \quad v_k(\eta) = v_k(\eta') \quad \text{for every } k.$$

The modified Cholesky representation is unique, so these equalities imply $\Sigma(\eta) = \Sigma(\eta')$.

For an off-diagonal pair (k, ℓ) , equation (22) gives

$$\Sigma_{k\ell}(\eta) = \frac{\sigma}{\sqrt{\lambda}} + \sigma^2 \min\{t_k, t_\ell\}. \quad (24)$$

By the update-time condition, select three update times $t_1 < t_2 < t_3$. For the pairs (1, 2) and (2, 3), the time terms in equation (24) are t_1 and t_2 , respectively. Applying equality of these two covariances under η and η' and subtracting yields

$$(\sigma^2 - \sigma'^2)(t_1 - t_2) = 0.$$

Because $t_1 \neq t_2$ and $\sigma, \sigma' > 0$, we obtain $\sigma = \sigma'$. Substitution into either equality in equation (24) then gives $\lambda = \lambda'$. Thus $\eta = \eta'$.

Because the Gaussian regression coefficients are now common, equality of the conditional means in equation (23) identifies the predictable locations recursively. For $k = 1$, it gives $d_1(\mu_0, r) = d_1(\mu'_0, r')$. If the locations agree through $k - 1$, equality of the k th conditional means gives $d_k(\mu_0, r) = d_k(\mu'_0, r')$. Hence, for every k ,

$$\mu_0 + \frac{A_k}{r} = \mu'_0 + \frac{A_k}{r'}. \quad (25)$$

It remains to identify r , μ_0 , and the two costs. Write the common positive effort kernel in equation (5) as ψ_t , so that the empirical leading-order system has

$$m_i(t) = \frac{\psi_t}{c_i}, \quad m_j(t) = \frac{\psi_t}{c_j}, \quad \tau_\ell(t) = r m_\ell(t), \quad \ell \in \{i, j\}.$$

Because $\theta > 0$, $\psi_t > 0$. When $c_i \neq c_j$, the difference $\tau_i(t) - \tau_j(t) = r\psi_t(1/c_i - 1/c_j)$ is nonzero and has a constant sign. The cumulative difference A_k is therefore strictly monotone in t_k . Applying equation (25) at two distinct update times and subtracting gives

$$(A_k - A_\ell) \left(\frac{1}{r} - \frac{1}{r'} \right) = 0.$$

Since $A_k \neq A_\ell$, we obtain $r = r'$, and substitution into equation (25) gives $\mu_0 = \mu'_0$.

Given r , the identified intensity difference determines $m_i - m_j = (\tau_i - \tau_j)/r$. Conditional on (η, μ_0) and the observed history, equation (15) then identifies the perceived-state path and hence ψ_t . Finally,

$$c_i = \frac{r\psi_t}{\tau_i(t)}, \quad c_j = \frac{r\psi_t}{\tau_j(t)},$$

so c_i and c_j are separately identified. If $c_i = c_j$, then $A_k = 0$ for every k , and a common rescaling of (r, c_i, c_j) leaves both submission intensities unchanged. Thus the full vector is not identified at equality and becomes weakly identified as the two costs approach one another.

We have shown that equality of the joint likelihood first implies $\eta = \eta'$ and then $\vartheta = \vartheta'$. Therefore Θ is locally identified at Θ_0 . $\square$

Appendix B: Auxiliary Results

B.1. Solve S_t in Equation (4)

If S_t is in steady state, $dS_t/dt = 0 \Leftrightarrow S_t = \bar{S} \equiv \sigma/\sqrt{\lambda}$. If S_t is not in steady state, i.e., $S_t \neq \bar{S}$, we first isolate the two variables and get

$$\frac{dS}{\sigma^2 - \lambda S^2} = dt$$

Integrating both sides gives

$$t = \int \frac{dS}{\sigma^2 - \lambda S^2} = \frac{1}{\sigma\sqrt{\lambda}} \int \frac{d(S\sqrt{\lambda}/\sigma)}{1 - (S\sqrt{\lambda}/\sigma)^2} \equiv \frac{1}{\sigma\sqrt{\lambda}} \int \frac{du}{1 - u^2},$$

where $u = S\sqrt{\lambda}/\sigma = S/\bar{S}$. For $\tanh^{-1}(u)$ we require $|u| < 1$ ($S < \bar{S}$), and for $\coth^{-1}(u)$ we require $|u| > 1$ ($S > \bar{S}$). Hence,

$$\sigma\sqrt{\lambda} \cdot t = \begin{cases} \tanh^{-1}(u) - K_1, & \text{if } |u| < 1 \\ \coth^{-1}(u) - K_2, & \text{if } |u| > 1 \end{cases} = \begin{cases} \tanh^{-1}(S/\bar{S}) - K_1, & \text{if } S < \bar{S} \\ \coth^{-1}(S/\bar{S}) - K_2, & \text{if } S > \bar{S} \end{cases}$$

Thus, in the non-steady-state case,

$$S = \begin{cases} \bar{S} \cdot \tanh(\sigma\sqrt{\lambda} \cdot t + K_1), & \text{if } S < \bar{S} \\ \bar{S} \cdot \coth(\sigma\sqrt{\lambda} \cdot t + K_2), & \text{if } S > \bar{S} \end{cases}$$

Finally, we determine the constants K_1, K_2 by the initial condition S_0 and have

$$K_1 = \tanh^{-1}(S_0/\bar{S}), \quad K_2 = \coth^{-1}(S_0/\bar{S})$$

If $\lambda = 0$, we have $S_t = S_0 + \sigma^2 t$, i.e., the estimation variance increases linearly over time. If $\lambda > 0$, the solution of (4) is

$$S_t = \begin{cases} \bar{S} \cdot \tanh \left\{ t \cdot \sigma\sqrt{\lambda} + \tanh^{-1}(S_0/\bar{S}) \right\} & \text{if } S_0 < \bar{S} \\ \bar{S} & \text{if } S_0 = \bar{S} \\ \bar{S} \cdot \coth \left\{ t \cdot \sigma\sqrt{\lambda} + \coth^{-1}(S_0/\bar{S}) \right\} & \text{if } S_0 > \bar{S} \end{cases}$$

Specifically, $\bar{S} = \sigma/\sqrt{\lambda}$ when $\lambda > 0$ and $\bar{S} = \infty$ when $\lambda = 0$.

B.2. Consistency Benchmark for σ^2

This appendix derives the private-leaderboard benchmark for σ used in Section 5.2. For a given contest, let $\Delta y_k = y_k - y_{k-1}$ denote the private-leaderboard gap increment between two adjacent update times, and let $h_k = t_k - t_{k-1}$ be the corresponding elapsed time. The posterior effort trajectories provide a plug-in drift adjustment

$$\hat{\mu}_k = \int_{t_{k-1}}^{t_k} [\hat{n}_i(s) - \hat{n}_j(s)] ds.$$

Under the diffusion approximation, the drift-adjusted increment satisfies

$$\Delta y_k - \hat{\mu}_k \sim \mathcal{N}(0, \sigma^2 h_k), \quad k = 1, \dots, n.$$

Treating these increments as approximately independent, the log-likelihood for σ^2 is, up to constants,

$$\ell(\sigma^2) = -\frac{n}{2} \log(\sigma^2) - \frac{1}{2} \sum_{k=1}^n \log h_k - \frac{1}{2\sigma^2} \sum_{k=1}^n \frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k}.$$

Differentiating with respect to σ^2 and setting the derivative equal to zero gives

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{k=1}^n \frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k} = 0.$$

Therefore,

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{k=1}^n \frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k}. \quad (26)$$

In the empirical application, $\hat{\mu}_k$ is computed from the Bayesian posterior effort paths, so this benchmark is a plug-in private-leaderboard consistency check rather than a fully model-free estimator. Moreover, asynchronous update timing means that adjacent displayed gaps may combine a new score for one contestant with a carried-forward score for the other. The expression in (26) is therefore an approximate benchmark for the carried-forward gap process, not an exact maximum likelihood estimator for synchronously observed latent diffusion increments.

Appendix C: Statistics

Table 9 reports raw contest-level estimates for transparency, even though some benchmark values are highly dispersed. Large values of λ^{bench} can arise mechanically when public-private score discrepancies are very small, because the benchmark is inverse-quadratic in those discrepancies. Accordingly, cross-contest comparisons of λ^{bench} are most interpretable on a transformed scale (as used in the main-text regressions).

Table 9: Contest-Level Structural Estimates and Benchmarks

Contest Id	N_i	N_j	Prize (100,000 USD)	Public Data %	Bayesian Posterior Mean						Diagnostic Benchmarks	
					λ	σ	μ_0	c_i	c_j	r	λ^{bench}	σ^{bench}
2445	19	7	0.0500	25	32.619	0.223	-0.191	0.657	1.544	13.001	79.821	0.127
2454	19	13	0.0015	62	8.547	0.704	4.295	0.058	0.094	119.189	38.095	0.570
2464	11	12	0.0095	20	2.836	1.665	-0.789	0.135	0.120	75.097	7.107	3.799
2478	11	13	0.0095	30	13.420	0.290	0.067	0.450	0.375	26.375	1397.214	0.231
2489	12	21	0.0050	10	8.433	0.265	0.275	0.335	0.182	41.075	125.491	0.588
2549	13	9	0.0050	30	3.434	0.709	-2.883	0.115	0.186	64.195	13.163	1.991
2551	49	30	0.0150	30	24.451	0.237	-0.017	0.273	0.445	29.884	5096.372	0.416
2667	66	54	0.3000	50	13.317	1.827	0.064	0.381	0.466	24.298	0.083	1.530
2749	18	11	0.0300	42	18.554	0.257	0.070	0.417	0.675	21.100	4872.050	0.591
2762	13	21	0.0100	35	7.339	0.227	0.802	0.304	0.184	43.702	4.110	0.370
2860	16	5	0.0100	30	25.762	0.155	-0.025	0.416	1.158	18.384	375.351	0.032
3064	13	5	0.0020	25	29.102	0.186	-0.001	0.134	0.419	44.423	421029.653	0.052
3080	24	27	0.0009	25	309.488	0.100	-0.471	0.121	0.112	78.903	16424.648	0.191
3288	20	10	0.0100	0	1.295	2.534	-0.366	0.080	0.187	80.707	0.096	1.783
3338	29	30	0.0150	30	90.042	0.189	-0.486	0.341	0.339	29.584	898.447	0.325
3353	31	6	0.1000	30	2.730	0.405	-1.327	0.499	2.154	12.647	846.158	1.075
3366	10	16	0.0050	50	9.932	0.508	2.385	0.218	0.123	58.530	273.117	0.858
3507	12	14	0.0050	33	8.469	0.474	-0.307	0.167	0.141	61.658	5.122	0.739
3509	8	13	0.0020	30	4.891	0.925	-0.049	0.175	0.090	77.067	85.308	0.965
3526	15	7	0.0500	20	19.071	0.293	0.631	0.762	1.457	11.308	185.860	0.185
3641	39	26	0.0150	43	23.257	0.186	0.728	0.334	0.497	26.529	33.952	0.313
3774	29	26	0.0050	50	4.893	0.328	-1.477	0.161	0.184	54.664	764.863	0.698
3800	34	15	0.0300	15	12.301	0.472	-2.890	0.264	0.613	25.668	57.667	0.794
3926	20	25	0.0030	25	301.643	0.097	-0.146	0.230	0.187	47.858	179.740	0.198
3928	15	21	0.0068	70	103.827	0.137	-0.580	0.407	0.300	30.667	4195.070	0.109
3929	36	31	0.0800	50	18.707	0.213	0.554	0.689	0.809	15.227	310.747	0.767
3960	47	49	0.0800	40	60.033	0.307	-1.233	0.523	0.521	20.892	56.933	0.454
4104	8	6	0.2000	20	8.848	0.686	0.023	1.217	1.540	9.217	1343.498	0.071
4366	31	30	0.0800	2	60.217	0.211	-0.122	0.647	0.719	16.199	299.806	0.375
4407	44	51	0.0400	19	1.713	0.553	0.506	0.329	0.273	34.991	526.875	1.991
4453	15	18	0.0200	30	12.435	0.461	1.085	0.460	0.378	25.144	718.111	0.456
4477	7	11	0.0200	50	8.998	0.706	-3.224	0.490	0.305	28.581	80.154	0.494
4488	11	8	0.0200	1	1.358	0.759	1.099	0.322	0.452	27.876	73.775	2.617
4493	19	14	0.1000	40	27.218	0.458	-0.424	0.684	0.932	14.739	328.980	0.380
4657	63	66	0.0400	30	257.543	0.138	0.006	0.605	0.584	18.900	61886.247	0.268
4699	13	22	0.0500	30	175.720	0.080	0.120	2.108	1.684	8.584	4805.379	0.024
4986	50	43	0.1000	50	63.403	0.149	-0.150	0.976	1.012	12.547	2637.968	0.282

Table 9 (continued)

Contest Id	N_i	N_j	Prize (100,000 USD)	Public Data %	Bayesian Posterior Mean						Diagnostic Benchmarks	
					λ	σ	μ_0	c_i	c_j	r	λ^{bench}	σ^{bench}
5056	18	34	0.0500	33	83.522	0.254	-0.981	0.883	0.508	18.711	30594.648	0.223
5144	15	37	0.2000	20	27.451	0.266	0.202	2.283	1.102	10.281	313.645	0.465
5174	42	47	0.0300	50	55.915	0.583	-0.909	0.248	0.221	41.041	17828.197	0.705
5229	28	31	0.1500	10	1.853	2.681	0.184	0.294	0.264	35.786	0.704	4.914
5261	36	45	0.1000	30	14.202	1.534	1.655	0.295	0.234	38.848	7473.741	1.332
5357	36	22	0.0500	30	38.662	0.322	0.743	0.445	0.704	20.411	415.700	0.395
5390	21	17	0.0400	30	11.970	0.302	0.055	0.466	0.593	21.625	102.418	1.042
5497	24	44	0.0400	30	148.564	0.135	-0.560	1.097	0.692	17.504	452226.832	0.300
6322	23	22	0.1000	66	45.899	0.203	-0.015	0.985	1.012	12.119	21008.033	0.367
6927	19	16	0.0400	1	69.137	0.206	-0.104	0.606	0.719	16.985	788.005	0.212
7115	19	38	0.0400	30	0.725	0.418	0.355	0.666	0.332	23.273	5212.948	1.839
7162	51	14	0.0100	50	2.731	0.449	-1.322	0.092	0.370	52.396	1648.108	1.359
7634	44	53	0.0200	2	8.190	0.709	-1.277	0.206	0.170	51.169	465.122	1.518
7878	5	7	0.0400	29	1.386	1.163	-0.548	0.808	0.591	16.816	5.532	1.685
8076	34	29	0.0600	10	17.709	0.257	-0.919	0.511	0.645	18.120	604.105	0.562
8078	29	32	0.0400	36	1.278	1.778	1.656	0.192	0.174	52.212	9.935	3.725
8219	16	23	0.0040	22	0.474	0.973	-0.080	0.117	0.076	94.485	21748.405	8.984
8396	53	59	0.0050	1	0.917	1.024	6.651	0.079	0.071	116.761	63.714	2.649
9120	69	46	0.1000	20	899.833	0.060	-0.159	1.773	1.963	9.647	1505.779	0.168
9949	21	35	0.0500	20	4.082	1.081	8.261	0.423	0.247	32.342	455.392	1.455
10200	22	35	0.0400	9	19.693	1.198	-7.557	0.303	0.186	40.903	1583.173	1.071
10684	16	58	0.0400	57	4.125	0.749	-7.145	0.626	0.166	31.481	1.460	2.462
13333	18	19	0.0200	25	270.719	0.098	-0.098	0.654	0.684	18.043	6184.689	0.057
14242	64	65	0.0300	20	71.133	0.208	0.057	0.382	0.375	27.741	20.646	1.121
14420	56	20	0.2000	22	2.609	1.053	-3.416	0.348	0.962	19.262	0.381	4.040
18045	40	68	0.0400	30	25.252	0.718	-3.158	0.348	0.202	36.667	10.708	0.754
19018	62	67	0.0800	30	13.137	0.410	1.430	0.471	0.442	23.798	1542.073	1.133
19991	40	11	0.0400	20	72.774	0.154	0.539	0.628	1.837	17.334	152.060	0.179
20270	20	16	0.0400	30	26.590	0.358	-0.455	0.422	0.528	23.259	12.643	0.385
21669	61	60	0.0100	21	14.026	0.595	2.483	0.124	0.126	72.455	165.216	1.460
22962	23	18	0.0400	24	5.015	0.497	-0.833	0.332	0.428	27.028	527.862	1.182
23249	21	40	0.0100	16	1512.555	0.033	-0.012	0.886	0.818	17.772	101330.854	0.039
23652	24	24	0.0100	20	9.473	0.409	-0.077	0.178	0.178	54.326	5150.468	1.614
37077	11	46	0.0400	24	5.428	0.516	0.122	0.969	0.231	22.854	259.940	0.835
38128	27	52	0.0800	42	30.523	0.404	-0.314	0.796	0.411	18.621	29.324	0.786
38760	10	17	0.0500	33	1.211	0.544	-0.021	0.659	0.384	21.435	342920.651	0.985

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